Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

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2. Study Design

This pivotal, phase 3 study was a multinational, multicenter, randomized, double-blind, placebo-controlled, parallel-group trial. The objective was to assess the effects of dupilumab administered subcutaneously (SC) for a 52-week treatment period in children aged 6 to <12 years with uncontrolled moderate-to-severe asthma. Dupilumab was administered as add-on therapy to a High-dose inhaled corticosteroid (ICS) alone, or Medium- or High-dose ICS in combination with a second controller (e.g., long-acting β2 agonist [LABA], long-acting muscarinic antagonist [LAMA], leukotriene receptor antagonists [LTRA], or methylxanthines).

The initial targeted study population consisted of children aged 6 to 11 years with uncontrolled asthma. In order to narrow the focus of the study to the population for which efficacy had been demonstrated in adults and adolescents (NCT02414854), and to be consistent with the approved label indications for adults and adolescents with asthma, two primary efficacy populations were evaluated: those with the type 2 inflammatory asthma phenotype defined as either baseline blood eosinophil count of ≥150 cells/µl or baseline fractional exhaled nitric oxide (FE_{NO}) ≥20 parts per billion (ppb), and the population with baseline blood eosinophil count ≥300 cells/µl. A total of 402 patients were planned to

be randomized, including approximately 345 patients with the type 2 inflammatory asthma phenotype and approximately 255 patients with baseline eosinophil count ≥300 cells/µl.

Patients were randomized in a 2:1 ratio to receive SC injection of dupilumab or placebo every 2 weeks (q2w). Patients with a body weight at randomization >30 kg received 200 mg (1.14 ml) dupilumab or matching placebo q2w, and patients with a body weight at randomization ≤30 kg received 100 mg (0.67 ml) dupilumab or matching placebo q2w. Due to the different volumes administered with the 100 mg versus 200 mg doses, two matching placebos were required for this study. Consequently, patients and Investigators were blinded to whether patients were receiving dupilumab or placebo, but not to the dose/volume of the injection (200 mg/1.14 ml; 100 mg/0.67 ml). After randomization, the assigned weight-tiered dose regimens of dupilumab at randomization were maintained during the 52-week treatment period of the study.

Randomization was stratified by ICS dose level (Medium, High) at screening, blood eosinophil count (<300 cells/µl and ≥300 cells/µl) at screening, and by region (Latin America, Eastern Europe, and Western countries). Dose levels considered to be Medium- or High-dose ICS in children aged 6 to 11 years were adapted from the Global Initiative for Asthma (GINA) 2015 guidelines that were applicable at the time of study initiation.

To ensure a representative asthma population in the pediatric age group, alerts were initially built into the interactive voice response system/ interactive voice response system to limit enrolling patients in the following stratification groups:

- Background therapy with Medium-dose ICS in no more than 50% (236) of patients; and
- Blood eosinophil count <150 cells/µl in no more than 20% (94) of patients.

The study consisted of three periods with a total duration of 68±1 weeks for each patient (Fig. S1): a screening period (4±1 weeks); a randomized double-blind treatment period (up to 52 weeks) during which patients received dupilumab or placebo administered as SC injections; and a post-treatment

period (12 weeks) for patients who did not participate in a 1-year long-term extension study, which was offered to patients who completed the randomized treatment period of the study. Home administration of the investigational medicinal product was allowed after week 12 at the approval of the investigator. Parents or legal guardians could administer the investigational medicinal product at home when clinic visits were not required.

Prior to and during the screening period, patients were required to be on a stable-dose background therapy of Medium-dose ICS with a second controller medication (i.e., LABA, LTRA, LAMA, or methylxanthines) or High-dose ICS alone or High-dose ICS with a second controller, for at least 3 months with a stable dose ≥1 month prior to screening visit 1. During the screening period, patients had to adhere to a background therapy, and non-compliance (<80% of total number of prescribed doses of background medication taken) was grounds for exclusion. During the randomized treatment period, patients continued taking their background controller medication(s) at the stable dose used during the screening period. Patients requiring a third controller medication for their asthma were not considered eligible for this study. Patients were allowed to use albuterol/salbutamol or levalbuterol/levosalbutamol as reliever medication as needed during the study.

For patients who experienced deterioration of asthma during the study (recorded as a loss of asthma control [LOAC] event), the ICS dose could be temporarily increased up to 4-fold for a maximum of 10 days, as indicated and upon recommendation of the physician and/or Investigator. Treatment could then be changed to systemic corticosteroids (SCS) (severe exacerbation event) or reverted back to the baseline ICS dose, depending on asthma symptom progression. Patients could be treated with SCS at any time and as clinically indicated based on the presence of symptoms consistent with a severe asthma exacerbation event, as per the Investigator's judgment. In addition, when a patient experienced two or more severe asthma exacerbation events anytime during the study, a permanent change (i.e., step up in Medium- to High-dose ICS or addition of second controller for patients on High-dose ICS monotherapy)

in their stable-dose background controller medication could occur, as indicated and upon recommendation of the Investigator.

3. Supplementary Methods

3.1 Eligibility Criteria

Inclusion criteria

- Children aged 6 to 11 years with a physician diagnosis of persistent asthma for ≥12 months prior to screening based on clinical history and examination, pulmonary function parameters according to GINA 2015 guidelines,[1] and the following criteria:
  - Existing background therapy of Medium-dose ICS with a second controller medication (i.e., LABA, LTRA, LAMA, or methylxanthines) or High-dose ICS alone or High-dose ICS with second controller, for at least 3 months with a stable dose ≥1 month prior to screening visit 1
  - Pre-bronchodilator forced expiratory volume in 1 second (FEV_1) ≤95% of predicted normal or pre-bronchodilator FEV_1 /forced vital capacity (FVC) ratio <0.85 at screening and baseline visits
  - Reversibility of at least 10% in FEV_1 after the administration of 200 to 400 µg (2 to 4 puffs with metered-dose inhaler [MDI]) of albuterol/salbutamol or 45 to 90 µg (2 to 4 puffs with MDI) of levalbuterol/levosalbutamol reliever medication before randomization (up to 3 opportunities during the same visit were allowed, with a maximum of 12 puffs of reliever medication if tolerated by the patient)
  - Must have experienced a severe exacerbation within 1 year prior to screening visit 1, defined as any of the following events:
    - Treatment with a SCS (oral or parenteral) prescribed by a healthcare professional for worsening asthma at least once, or
    - Hospitalization or emergency medical care visit for worsening asthma

- Evidence of uncontrolled asthma, with at least one of the following criteria during the 4(±1)-week screening period:
  - 5-item Asthma Control Questionnaire – Interviewer-Administered (ACQ-5-IA) score ≥1.5[2,3] on at least 1 day of the screening period including visit 2
  - Use of reliever medication (i.e., albuterol/salbutamol or levalbuterol/levosalbutamol), other than as a preventive for exercise induced bronchospasm, on 3 or more days/per week on at least 1 week during the screening period
  - Sleep awakening due to asthma that required the use of reliever medication at least once during the screening period
  - Asthma symptoms 3 or more days/week on at least 1 week during the screening period

- Willing and able to comply with clinic visits and study-related procedures
- With parent(s)/caregiver(s)/legal guardian(s) able to understand the study requirements
- Patients aged ≥6 years (or above an age determined by the Institutional Review Board [IRB]/Independent Ethics Committee [IEC] and in accordance with the local regulations and requirements) must provide written informed assent, and their parent(s)/caregiver(s)/legal guardian(s) must provide written informed consent
- Patients/parent(s)/caregiver(s)/legal guardian(s), as appropriate, were able to understand and complete study-related questionnaires

Exclusion criteria

Exclusion criteria related to study methodology

- Patients aged <6 or ≥12 years
- Patients <16 kg body weight
- Any other chronic lung disease (cystic fibrosis, bronchopulmonary dysplasia, etc.) that may impair lung function

- Baseline blood eosinophil counts of >1500 cells/µl at screening

3.2 Data and Safety Monitoring Committee

The data and safety monitoring committee consisted of top leading experts in areas of expertise, including but not limited to: Pediatric Allergy and Immunology, Dermatology and Epidemiology, Internal Medicine, Allergy, Asthma and Immunology, Dermatology, Epidemiology, and Large-Scale Data Review.

3.3 Narrative of Eosinophilia Case Associated with Clinical Symptoms

Treatment-emergent adverse events (TEAEs) of eosinophilia occurred in 18 patients in the dupilumab group, 16 of these cases were self-limiting laboratory findings without any associated symptoms. Two events of eosinophilia were considered serious adverse events (SAEs), both in the dupilumab group. One of them was associated with clinical symptoms, a description of this one case is below:

Female patient in the group aged 9 to 11 years

A female patient in the aged 9 to 11 years group with last asthma exacerbation approximately 1 month prior to randomization requiring a 9-day treatment with an oral corticosteroid (OCS) developed asymptomatic eosinophilia on day 15, which was moderate in intensity. Baseline blood eosinophil count at screening was 10 cells/µl at screening. Eosinophil count was 3090 cells/µl on day 15 (vs. 10 cells/µl on day −28) and then 4610 cells/µl on day 45, and 5340 cells/µl on day 72. On day 73, the patient complained of dizziness (spells of blurred vision that would last a few seconds) and headache that was constant, frontal, and relieved by non-steroidal anti-inflammatory drugs (NSAIDs). On day 74, 2 days after the sixth investigational medicinal product (IMP) injection, the patient reported eosinophilia of severe intensity and that was serious (hospitalization). Diagnosis of hypereosinophilia syndrome was ruled out based on electrocardiogram (ECG), transthoracic echocardiogram, magnetic resonance imaging (MRI) of the brain, and lumbar puncture (cerebrospinal fluid), which were all normal. The study

code was broken and treatment with the IMP was permanently discontinued. The patient recovered from blurred vision and headache (day 110) and from eosinophilia (day 284). The Investigator assessed the events of headache, blurred vision, and eosinophilia as related to the IMP.

3.4 Narratives of parasitic infection cases

Parasitic infections were reported in 7 (2.6%) dupilumab-treated and one (0.7%) placebo-treated patient. Most frequent among these was enterobiasis, occurring in 5 (1.8%) dupilumab-treated patients. One (0.4%) dupilumab patient experienced ascariasis, one (0.4%) dupilumab patient experienced lice infestation and one (0.7%) placebo patient experienced parasitic gastroenteritis. Four of the six patients with enterobiasis/ascariasis were from Argentina or Poland; in three of six cases, no confirmatory test was performed, and diagnosis of parasitic infection was patient/parent reported. None of the parasitic infections were severe, all patients recovered after treatment with an anthelminthic, and none led to permanent discontinuation of the investigational medicinal product. A majority of these cases occurred in endemic regions.

A male patient in the 6-8 years age group

A male patient (dupilumab 100 mg q2w), in the 6-8 years age group had a medical history of asthma exacerbation 3 months prior to randomization and that was controlled with 7-day use of OCS, multiple atopic conditions, and a previous history of ascariasis and infection with candida glabrata. The patient experienced asymptomatic eosinophilia on Day 84 (4.33 Giga/L versus 0.72 Giga/L at baseline) and Day 168 (4.5 Giga/L). On Day 181, the patient was diagnosed with enterobiasis (moderate) after his mother noticed pinworm around the anus (no investigations performed). No action was taken with the IMP. The patient recovered from enterobiasis on Day 182 after corrective treatment with oral pyrantel and from eosinophilia on Day 224. Both events of enterobiasis and eosinophilia were assessed by the Investigator as not related to the IMP.

A female patient in the 6-8 years age group

A female patient (dupilumab 100 mg q2w), in the 6-8 years age group, with a medical history of 12 episodes of asthma exacerbation in the previous year requiring multiple hospitalizations and approximately 60 days of OCS use during that period prior to randomization, was diagnosed with enterobiasis (mild intensity) on Day 311 after she complained of anal itching and parasites were observed in stool by the study participant's mother. No other diagnostic test was performed. The patient recovered on Day 335 after corrective treatment with a single dose of mebendazole. The patient experienced asymptomatic eosinophilia (moderate intensity) on Day 364 (4.62 Giga/L versus 1.31 Giga/L at baseline). No action was taken with the IMP. The patient recovered on Day 392 (eosinophil count not provided). Both events of enterobiasis and eosinophilia were considered by the Investigator as not related to the IMP.

A female patient in the 9-11 years age group

A female patient (dupilumab 200 mg q2w), in the 9-11 years age group, was diagnosed with enterobiasis (moderate intensity) on Day 179, based on parasite test and culture. Prior to this on Day 170, the patient developed a nonserious event of eosinophilia (mild intensity) with eosinophil count of 1.4 Giga/L (versus 1.42 Giga/L at baseline), following which, a stool test was requested and was positive for oxiuriasis. Corrective treatment included mebendazole followed by tinidazole. The eosinophil count on Day 253 was 1.94 Giga/L. The patient recovered from both events. Treatment with IMP was continued. Both events were considered by the Investigator as not related to the IMP.

A female patient in the 9-11 years age group

A female patient (dupilumab 100 mg q2w) in the 9-11 years age group, was diagnosed with enterobiasis (mild intensity) on Day 87, based on clearly visible parasites in the stool (no diagnostic test performed). Eosinophil count was 2.54 Giga/L on Day 86 (versus 1.08 Giga/L at baseline). Treatment with IMP was continued. The patient recovered from enterobiasis on Day 169 after corrective treatment with 3 doses of mebendazole over 3 days. The eosinophil count was 1.2 Giga/L on the same day and 0.88 Giga/L on Day 253. The event was considered by the Investigator as not related to the IMP.

Patient (dupilumab 100 mg q2w):

Patient (dupilumab 100 mg q2w): this patient reported enterobiasis in addition to eosinophilia associated with clinical signs.

A female patient in the 6-8 years age group

A female patient (dupilumab 100 mg q2w) in the 6-8 years age group, experienced ascariasis on Day 371, 14 days after the last IMP administration. The diagnosis was based on a positive serum IgG for Ascaris lumbricoides. The eosinophil count was 3.59 Giga/L (0.26 Giga/L at baseline). The patient was treated with albendazole and recovered within 14 days. The event was considered by the Investigator as not related to the IMP.

A female patient in the 9-11 years age group

A female patient (dupilumab 200 mg q2w) in the 9-11 years age group, experienced a nonserious AE of pediculosis (lice infestation) on Day 422. The eosinophil count was 0.35 Giga/L at baseline. The patient was treated with permethrin topical application and recovered within 28 days. The event was considered by the Investigator as not related to the IMP.

A female patient (placebo group) in the 6-8 years age group

A female patient in the 6-8 years age group, was diagnosed with parasitic gastroenteritis on Day 153. The patient reported anal pruritus and the patient's mother reported to have found parasites in the patient's feces. The patient was empirically medicated without a confirmatory diagnostic test. The patient recovered within 38 days. The event was considered by the Investigator as not related to the IMP.

3.5 3.5 Severe exacerbations requiring hospitalization

Four of 271 dupilumab-treated patients and none of the 134 placebo-treated patients experienced SAEs of asthma exacerbation that required hospitalization.

All SAEs of asthma exacerbation were of moderate intensity in 3 patients and severe in 1 patient who had concurrent pneumonia that required hospitalization for 7 days. All 4 patients had trigger factors for asthma exacerbation: concomitant infection (TEAEs of pneumonia and otitis media), concomitant TEAE of upper respiratory tract infection, infection reported as a trigger factor, and changes in temperature and exercise reported as trigger factors. None of these events resulted in permanent treatment discontinuation and none of the events were considered by the Investigator to be related to the IMP. All 4 patients were ADA negative and all recovered. All 4 patients had their lung function improved during the course of the study. Three of the 4 patients reported at least 5 severe asthma exacerbations in the prior year, and the number of severe exacerbations observed during the course of the study as lower thanin the prior year (1 patient with 17 severe exacerbations the prior year had 4 events and 2 patients with 5 severe exacerbations the prior year had 1 exacerbation event during the 52-week treatment period).

3.6 Statistical analysis

The study was sufficiently powered to detect exacerbation reductions in the two primary populations (type 2 phenotype and baseline blood eosinophils ≥300 cells/µL). Sample size was estimated based on the observed effect sizes for these populations in the adult and adolescent phase III study.[4] With a randomization ratio of 2:1, and assuming placebo annualized severe exacerbation rates of 0.7 and 0.8, a sample of approximately 345 and 255 patients would have approximately 94% and 96% power to detect a 54% and 60% reduction in the annualized rate of severe exacerbations in the two primary populations, respectively, with a 2-tailed significance level of 0.05 (assuming an average exposure duration of 0.9 years and a dispersion parameter of 1.5).

Data were analyzed on an intent-to-treat population according to the assigned intervention, whether or not an intervention was received. All analyses included in this manuscript were prespecified unless specifically stated otherwise. The annualized rate of severe exacerbation events was analyzed using a negative binominal regression model that includes the total number of events observed from randomization up to week 52 or last study contact date (whichever comes earlier) as the response variable, and includes treatment group, age, baseline weight group, region, baseline eosinophil level, baseline FE_{NO} level, baseline ICS dose level, and number of severe exacerbation events within 1 year prior to the study as covariates, and log-transformed standardized observation duration as an offset variable. Change from baseline in percentage of predicted FEV_1 was analyzed using a mixed-effect model with repeated measures approach, including treatment, baseline weight, region, ethnicity, baseline eosinophil level, baseline FE_{NO} level, baseline ICS dose level, visit, treatment-by-visit interaction, baseline percentage of predicted FEV_1 value, and baseline-by-visit interaction as covariates. Changes from baseline in biomarker levels are presented using descriptive statistics unless otherwise specified.

To control the family-wise type I error for the statistical analyses of the primary and selected secondary efficacy end points, a step-down hierarchical testing procedure was applied at a two-sided 5%

significance level. Two separate testing orders were pre-specified, reflecting regional differences in the dupilumab indication label for adults and adolescents with asthma (Table S2 in the Supplementary Appendix).

The sample size of this study was based on a comparison between dupilumab versus placebo with regard to the primary end point of annualized rate of severe exacerbations over 52 weeks of treatment for the three populations of interest: patients with type 2 phenotype, patients with baseline blood eosinophils ≥300 cells/µl, and patients with baseline blood eosinophils ≥150 cells/µl, assuming the number of severe exacerbations follows a negative binomial distribution with a dispersion parameter of 1.5, an average exposure of 0.9 years and a randomization ratio of 2:1. For patients with the type 2 inflammatory phenotype, assuming a placebo annualized severe exacerbation rate of 0.7, with approximately 345 patients randomized, the study would have approximately 94% power to detect a 54% relative risk reduction in the annualized rate of severe exacerbations (with a 2-tailed test at an alpha level of 0.05). For patients with baseline blood eosinophils ≥300 cells/µl, assuming a placebo annualized severe exacerbation rate of 0.8, with approximately 255 patients randomized, this study would have approximately 96% power to detect a 60% relative risk reduction (i.e., annualized rate of 0.32 for the dupilumab group) in the annualized rate of severe exacerbations (with a 2-tailed test at an alpha level of 0.05).

For patients with baseline blood eosinophils ≥150 cells/µl, assuming a placebo annualized severe exacerbation rate of 0.7, and a dispersion parameter of 1.5, with approximately 327 patients randomized (218 for dupilumab and 109 for matching placebo group), the study will have approximately 93% power to detect a 54% relative risk reduction (i.e., annualized rate of 0.322 for the dupilumab group) in the annualized rate of severe exacerbations at the 2-tailed significance level of $\alpha = 0.05$ among these patients.

All efficacy end points were analyzed in the two primary efficacy analysis populations: the type 2 inflammatory asthma phenotype (i.e., baseline blood eosinophils ≥150 cells/µl or baseline FE_{NO} level ≥20 ppb) and the population with baseline blood eosinophils ≥300 cells/µl. In addition, the primary end point (annualized rate of severe exacerbation events), key secondary end point (percent of predicted FEV_1), and select secondary end points (change from baseline in 7-item Asthma Control Questionnaire – Interviewer-Administered [ACQ-7-IA] at week 24 and change from baseline in FE_{NO} at week 12) were analyzed in the population of patients with eosinophils ≥150 cells/µl, the population with baseline FE_{NO} ≥20 ppb, and in the intention-to-treat (ITT) population.

The annualized rate of severe exacerbation events was analyzed using a negative binominal regression model including the total number of events for each patient that occurred during the 52 weeks as the response variable, and with treatment group, age, weight (≥30 kg, >30 kg), region, baseline eosinophil level (<300 cells/µl, ≥300 cells/µl), baseline FE_{NO} level (<20 ppb, ≥20 ppb), baseline ICS dose level (Medium/High), and number of severe asthma exacerbation events prior to the study as covariates. Severe exacerbations were considered separate exacerbations if the interval between their start dates was equal to or greater than 28 days, meaning that a maximum of 13 exacerbations could be observed in a single participant during the 52-week study period.

Change from baseline in percentage of predicted pre-bronchodilator FEV_1 was analyzed using a mixed-effects model with repeated measures approach including treatment, baseline weight (≤30 kg, >30 kg), region, ethnicity, baseline eosinophil level (<300 cells/µl, ≥300 cells/µl), baseline FE_{NO} level (<20 ppb, ≥20 ppb), baseline ICS dose level (Medium/High), visit, treatment-by-visit interaction, baseline percentage of predicted FEV_1 value, and baseline-by-visit interaction.

Change from baseline in patient-reported outcomes were analyzed using a mixed-effects model with repeated measures approach, including treatment, age, baseline weight (≤30 kg, >30 kg), region,

baseline eosinophil level, baseline FE_{NO} level, baseline ICS dose level, visit, treatment-by-visit interaction, baseline value of the endpoint, and baseline-by-visit interaction as covariates.

Change from baseline in continuous end points such as percentage of predicted FEV_1 and patient-reported outcomes were analyzed using a mixed-effects model with repeated measures approach, including treatment, age, weight (≥30 kg, >30 kg), region, baseline eosinophils (<300 cells/µl, ≥300 cells/µl), baseline FE_{NO} level (<20 ppb, ≥20 ppb), baseline ICS dose level (Medium/High), visit, treatment-by-visit interaction, corresponding baseline value, and baseline-by-visit interaction. For end points analyzed in the population with baseline blood eosinophils ≥300 cells/µl and the population with baseline FE_{NO} ≥20 ppb, the covariates of baseline eosinophil level and baseline FE_{NO} level, respectively, were removed from the models. Time-to-event variable (time to first exacerbation and time to first LOAC) were analyzed using a Cox regression model with time-to-event as the dependent variable, and treatment, age, weight (≤30 kg, >30 kg), region, baseline eosinophil level (<300 cells/µl, ≥300 cells/µl), baseline FE_{NO} level (<20 ppb, ≥20 ppb), baseline ICS dose level (Medium/High) and number of severe asthma events prior to the study as covariates. The Kaplan–Meier method was used to derive the proportion of patients with a severe asthma exacerbation event at weeks 12, 24, 36, and 52 specific to each treatment group. All analyses were conducted using SAS software, version 9.4.

To control the type-I error rate for the analysis of efficacy end points, a hierarchical testing procedure was applied at a 2-sided 5% significant level (i.e., each hypothesis was formally tested only if the preceding one was significant at the 2-sided 5% level). Two hierarchical testing procedures were used based on the asthma indications in the approved labels in adults and adolescents for the US (baseline blood eosinophils ≥300 cells/µl) and the EU (baseline blood eosinophils ≥150 cells/µl or baseline FeNO ≥20 ppb). The same end points were analyzed in both hierarchies but in a different order (see Table S2).

For the primary endpoint, if patients withdrew from the study before the final visit, severe exacerbation events that may have occurred after discontinuation could not be observed. These patients were considered as patients with missing data for the endpoint. For the key secondary endpoint, if patients withdrew from the study before week 12, percentage of predicted FEV1 data was missing. No imputation was performed for missing values in the analysis.

Sensitivity analyses in the two primary populations were conducted to assess the robustness of the conclusion drawn based on the main model for the primary and key secondary endpoint analyses to the missing data (see Table S3-5; Fig S2 and S3):

- Primary endpoint sensitivity analyses:
    - Pattern mixture model-multiple imputation (PMM-MI)

For each patient with partial missing data of severe exacerbation events, the unobserved number of events after study discontinuation up to Week 52 was imputed on monthly basis. Since the definition of severe exacerbation required the start dates of 2 consecutive severe exacerbation events experienced by one patient to depart at least 28 days from each other, each patient is possible to have a maximum of one event per month that is defined as 28 days. A logistic regression model therefore could be used to impute whether a patient had an event or not for each of the corresponding unobservable months. This imputation was repeated multiple times. A final statistical inference was obtained by using Rubins' formula to combine the binomial regression results based on each of the imputed datasets.

    - Control based PMM-MI

Similar to aforementioned PMM model except that the individual monthly mean probability was calculated based on observations in placebo arm only in the logistic regression, with adjustment of missing observation duration.

    - Tipping point analysis

Similar to aforementioned PMM model except that if a patient with missing data was on dupilumab, the odds in the logistic model for each unobservable month were increased, and vice versa if the patient was on placebo, the odds were decreased. This adjustment artificially lowers the predicted event rates in placebo group and increases the predicted event rates in Dupilumab group. For each increasing/decreasing ratio, a p-value was obtained based on the imputed dataset with this specific ratio. A sequence of different ratios was explored until p-value became >0.05.

The missing data sensitivity analyses for the primary endpoint analysis was performed in the Type 2 inflammatory asthma phenotype and baseline blood eosinophils ≥0.3 Giga/L populations. For each of the above methods, a negative binomial model was fitted using the complete dataset composed of observed and imputed data, including the total number of observed and imputed events during the 52 weeks as the response variable. When performing the sensitivity analyses in the population with Type 2 inflammatory asthma phenotype, the model included the treatment group, age, baseline weight (≤30kg, >30kg), region (pooled country), baseline eosinophil level, baseline FeNO level, baseline ICS dose level and number of severe exacerbation events within 1 year prior to the study as covariates. When performing the sensitivity analyses in the population with baseline blood eosinophils ⩾0.3 Giga/L, the baseline eosinophil level was removed from the model covariates. Log transformed observation duration was the offset variable.

- Key secondary endpoint sensitivity analyses:
    - Pattern mixture model-multiple imputation (PMM-MI)

Missing % predicted pre-BD FEV1 values were imputed multiple times with adjustment for covariates including treatment, region, ethnicity, baseline eosinophil level, baseline FeNO level, baseline ICS dose level, baseline % predicted FEV1 value and reason of treatment discontinuation. Each of the complete

datasets was analyzed using the ANCOVA model with change from baseline in % predicted pre-BD FEV1 at Week 12 as the response variable, treatment, baseline weight, region, ethnicity, baseline eosinophil level, baseline FeNO level, baseline ICS dose level, and baseline % predicted pre-BD FEV1 value as covariates. Then the SAS MIANALYZE procedure was used to generate statistical inference by combining results using Rubin's formula.

- Control based PMM-MI

The analysis was similar as the standard PMM-MI described above, except that it assumed that after withdrawal from the study, patients from the dupilumab groups would exhibit the same future evolution of % predicted pre-BD FEV1 as patients on the placebo who were not exposed to dupilumab.

- Tipping point analysis

First, missing values was imputed by PMM-MI as illustrated above. The imputed values in placebo group were then shifted by adding a sequence of positive values and the imputed values in dupilumab group will be shifted by subtracting a sequence of positive values. For each combination of the shift parameters, each of the imputed and shifted datasets was analyzed with the ANCOVA model and their results were combined using Rubin's formula to generate statistical inference. LS mean difference between dupilumab and placebo in change from baseline in % predicted pre-BD FEV1 at Week 12 and the corresponding p-values were provided for each combination of shift parameters.

The missing data sensitivity analyses for the key secondary endpoint analysis were performed in the Type 2 inflammatory asthma phenotype and baseline blood eosinophils ⩾0.3 Giga/L populations. When performing the sensitivity analyses in the population with Type 2 inflammatory asthma phenotype, the model included treatment, baseline weight, region, ethnicity, baseline eosinophil level, baseline FeNO level, baseline ICS dose level, visit, treatment by-visit interaction, baseline % predicted FEV1 value and baseline-by-visit interaction as covariates. When performing the sensitivity analyses in the population

with baseline blood eosinophils ⩾0.3 Giga/L, the baseline eosinophil level was removed from the model covariates.

The Mixed Model for Repeated Measures (MMRM) was pre-specified to analyze the repeated measures of the continuous endpoint (such as FEV1, FEV1PP). Although called MMRM, this model is essentially the general (Gaussian) linear model (e.g.[5,6]), which can be viewed as the marginal model for the fixed effects in a general linear mixed-effects model with assuming an unstructured variance-covariance matrix for the random errors to account for the random effect. The naming MMRM is mostly a tradition in clinical studies. In the current study, the MMRM can be expressed as

$$y_n \sim N(X_n \beta_0, \Sigma_n),$$

where $\{y_n : n = 1, \dots, N\}$ are the observed outcome vector (y_n is a $p_n \times 1$ vector) from N independent subjects, X_n is the $p_n \times k$ dimensional design matric for subject n, $\beta_0 = (\beta_{10}, \dots, \beta_{k0})'$ is the true parameter vector, and Σ_n is the true variance-convariance matrix of y_n. Details can be found in references Fan et al.[5] and Harville DA.[6]

3.7 Overview of efficacy endpoints

Primary efficacy variable

The primary endpoint for this study was the annualized rate of severe exacerbation events during the 52-week placebo-controlled treatment period, defined as the number of severe exacerbation events with onset during the 52-week treatment period per patient-year. Data were collected for a 52-week period to account for seasonality.

Secondary efficacy variables

The key secondary endpoint was the change from baseline in pre-bronchodilator (pre-BD) percent predicted FEV1 at Week 12. Week 12 was chosen to align with FEV1 outcomes in the phase 3 studies in adolescents/adults.

Other secondary endpoints were as follows:

- Change from baseline in percent predicted pre-BD FEV_1 at Weeks 2, 4, 8, 24, 36, and 52 and other time points assessed.
- Time to first severe exacerbation event during 52-week treatment period.
- Time to first LOAC event during 52-week treatment period.
- Change from baseline in other lung function measurements (absolute and relative FEV_1, AM/PM peak expiratory flow, FVC, forced expiratory flow (FEF) 25-75%, post-bronchodilator percent predicted FEV1) at Weeks 2, 4, 8, 12, 24, 36, 52, and other time points assessed.
- Effect of dupilumab on healthcare resource utilization (HCRU) — not presented here.
- Change from baseline at Weeks 2, 4, 8, 12, 24, 36, and 52 and other timepoints in:
    - Morning/evening asthma symptom score (electronic diary) — not presented here.
    - ACQ-7-IA and ACQ-5-IA — ACQ-5 not presented here. Week 24 was chosen to align with asthma control outcomes in the phase 3 studies in adolescents/adults.
    - Use of reliever medication and of SCS

  - Number of asthma-related nocturnal awakenings requiring reliever medication use — *not presented here*

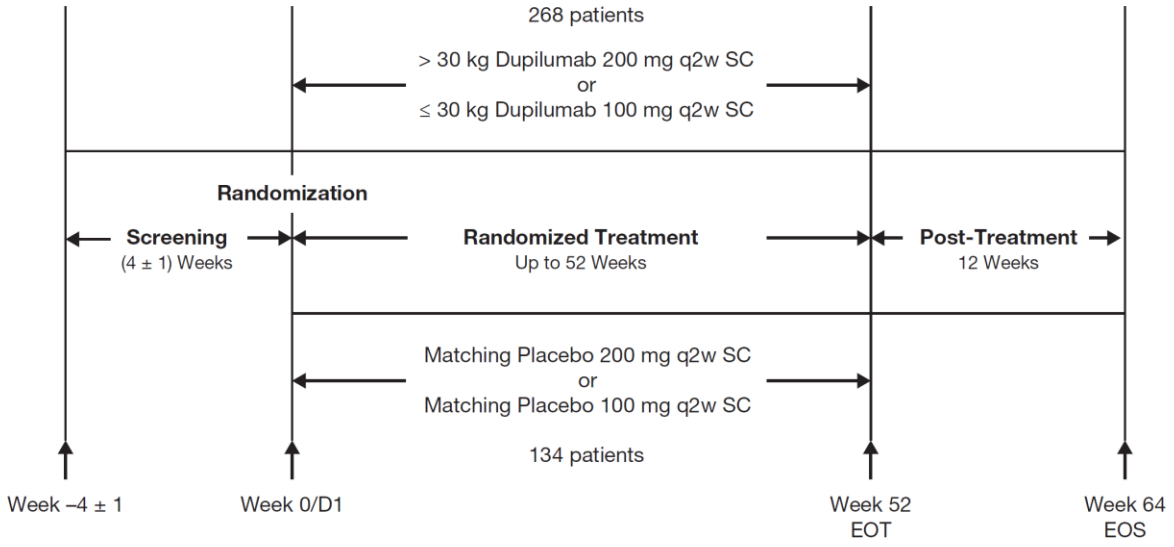
- Change from baseline at Weeks 12, 24, 36, 52, 64 in Pediatric Asthma Quality of Life Questionnaire with Standardized Activities – Interviewer Administered (PAQLQ(S)-IA) score, for children ≥7 to <12 years old at randomization Visit 2.
- Change from baseline in fractional exhaled nitric oxide (FeNO) at Week 12.

Other exploratory endpoints were as follows:

- Change from baseline and blood biomarkers (thymus and activation regulated chemokine [TARC], serum total immunoglobulin E [IgE]).
- Proportion of patients requiring a permanent step up in background controller medication after 2 or more severe asthma exacerbation events.
- Change from baseline to Week 52 in the following patient or caregiver reported outcomes:
  - Pediatric Rhinoconjunctivitis Quality of Life Questionnaire – Interviewer Administered (PRQLQ-IA) score, in children with history of allergic rhinitis — not presented here;
  - EuroQol-5 Dimension Youth instrument (EQ-5D-Y) score and visual analogue scale (EQ-VAS) — not presented here
  - Pediatric Asthma Caregiver's Quality of Life Questionnaire (PACQLQ) for caregivers of children ≥7 years old— not presented here.
- Slope of percent predicted post-BD FEV1— not presented here.

4 Tables and Figures

Figure S1. VOYAGE Study Design.



D denotes day, EOS end of study, EOT end of treatment, ICS inhaled corticosteroid, q2w every 2 weeks and SC subcutaneously.

Figure S2. Sensitivity analysis: Forest plot of relative risk in annualized severe exacerbations by analysis methods

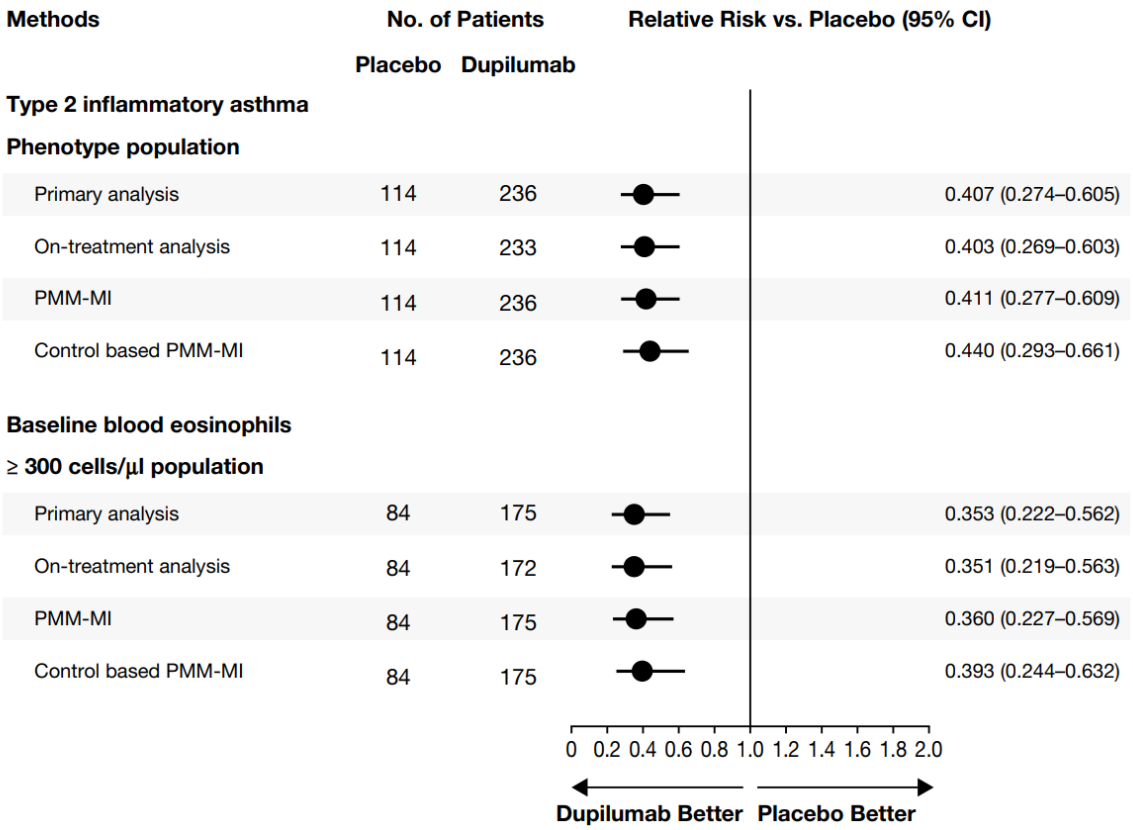


Figure S3. Sensitivity analysis: Forest plot of summary of change from baseline in pre-bronchodilator % predicted FEV_1 at Week 12 by analysis methods

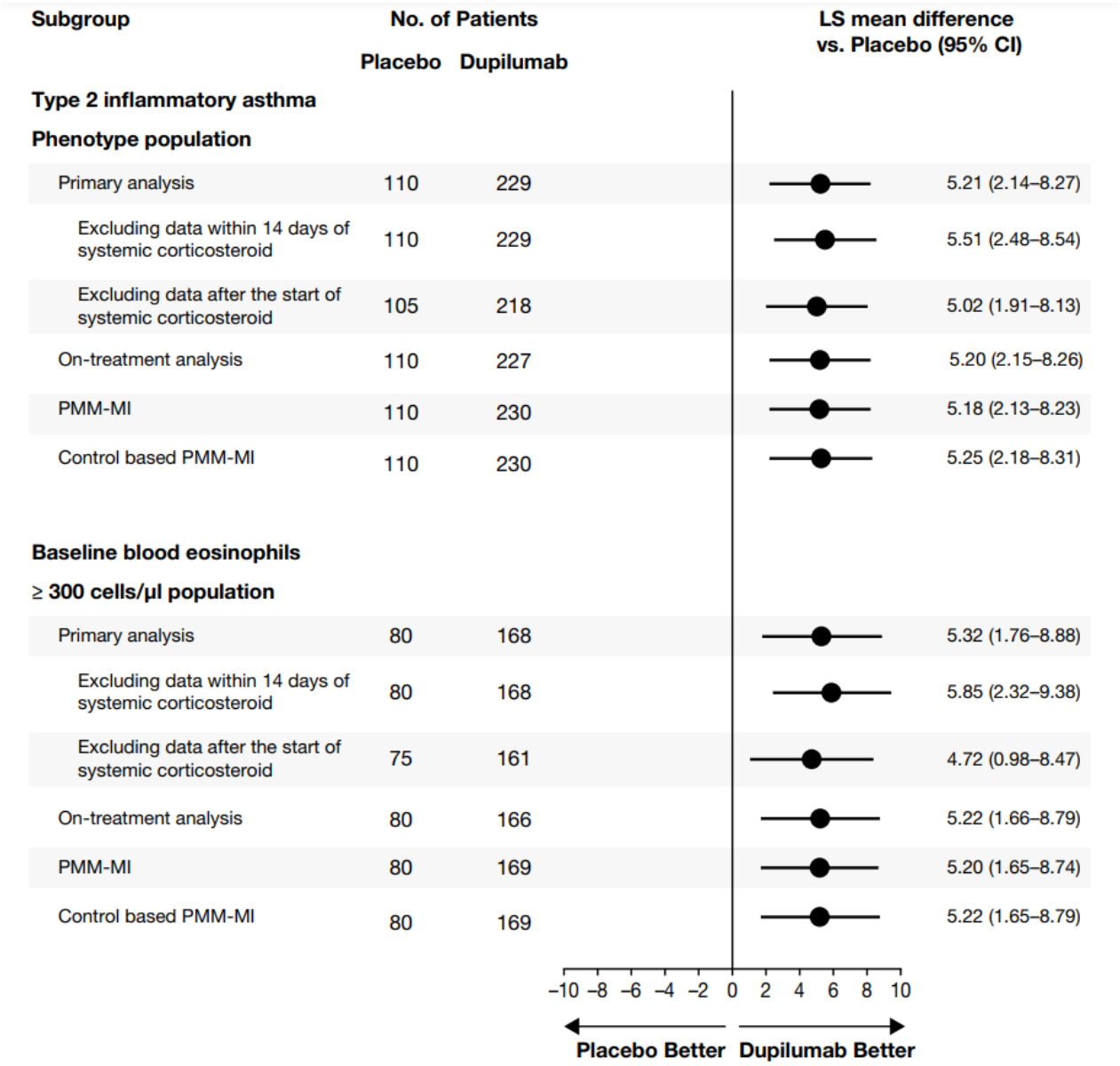
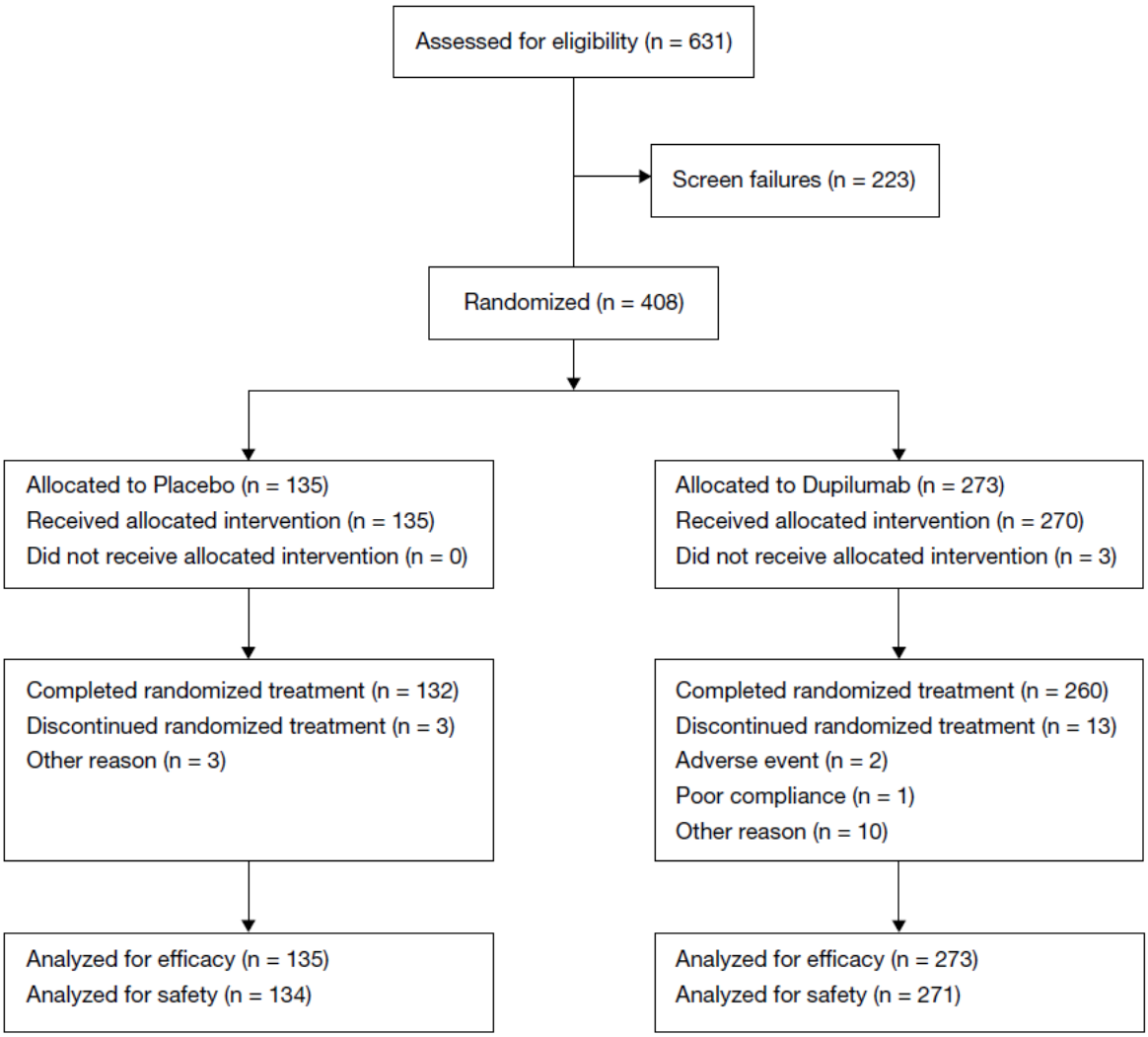


Figure S4. CONSORT Diagram of Patient Disposition.



*The leading reason for screen failure was not meeting the main inclusion criterion (lack of appropriate background therapy, exceeding the maximum pre-bronchodilator FEV_1 at baseline, FEV_1 reversibility under the specified minimum, or insufficient evidence for uncontrolled asthma)

Figure S5. Forest Plots of the Risk of Severe Asthma Exacerbations in Patients Receiving High-Dose ICS (Populations Defined by Baseline Blood Eosinophils and FE_{NO} Levels and Overall ITT Population).

CI denotes confidence interval, FE_{NO} fractional exhaled nitric oxide, ICS inhaled corticosteroid, ITT intention-to-treat, and ppb parts per billion.

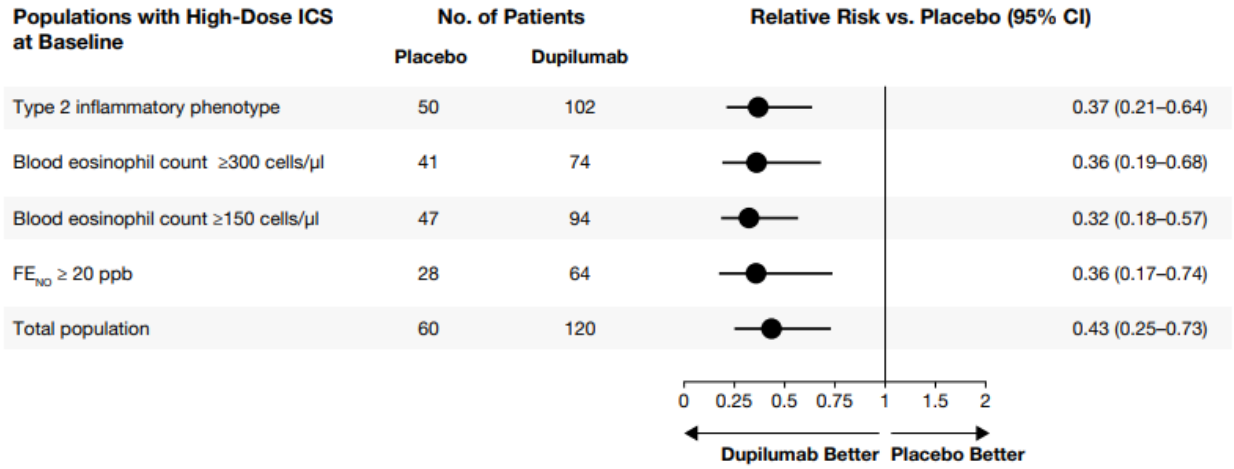
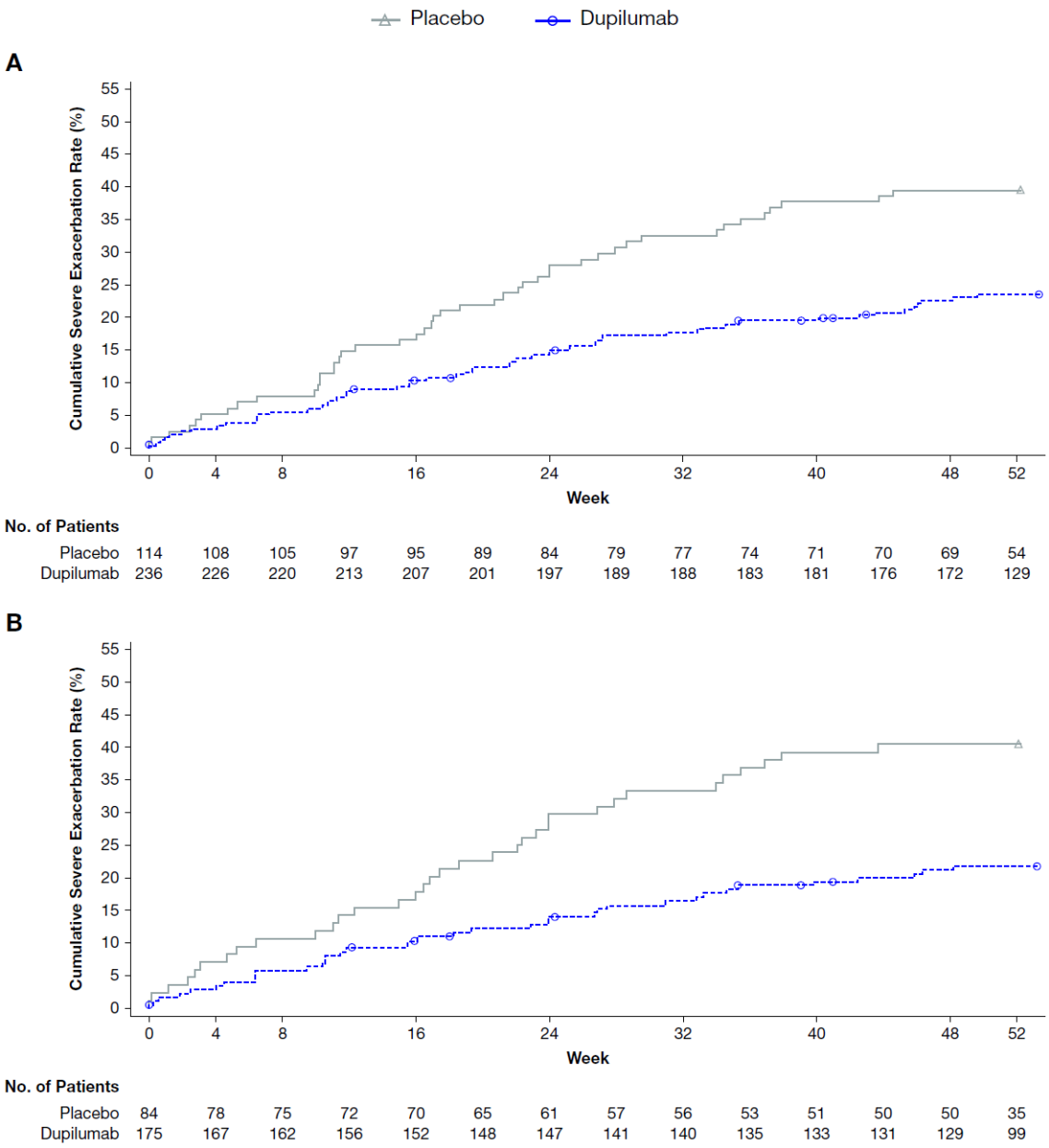
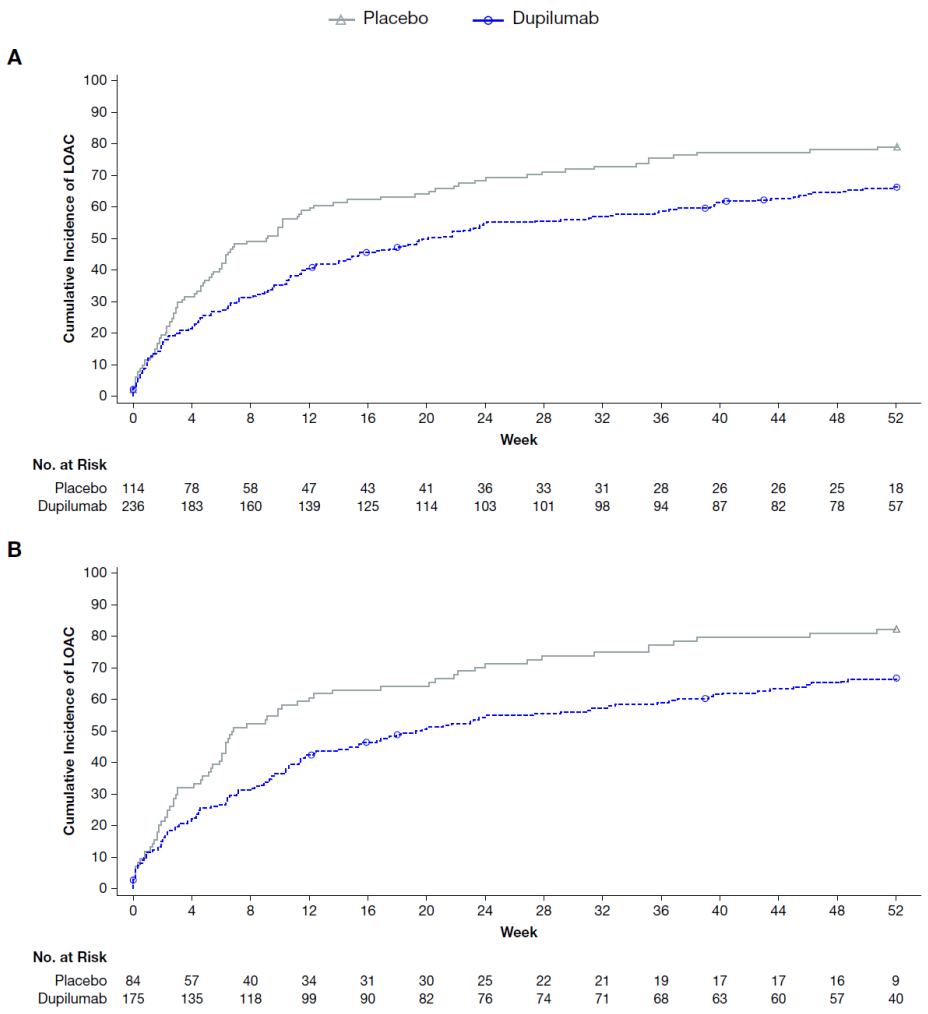


Figure S6. Kaplan–Meier Plot of Time to First Severe Exacerbation Event During the 52-Week Treatment Period in the Populations with the Type 2 Inflammatory Asthma Phenotype (A) and with Blood Eosinophils ≥300 cells/µl (B).



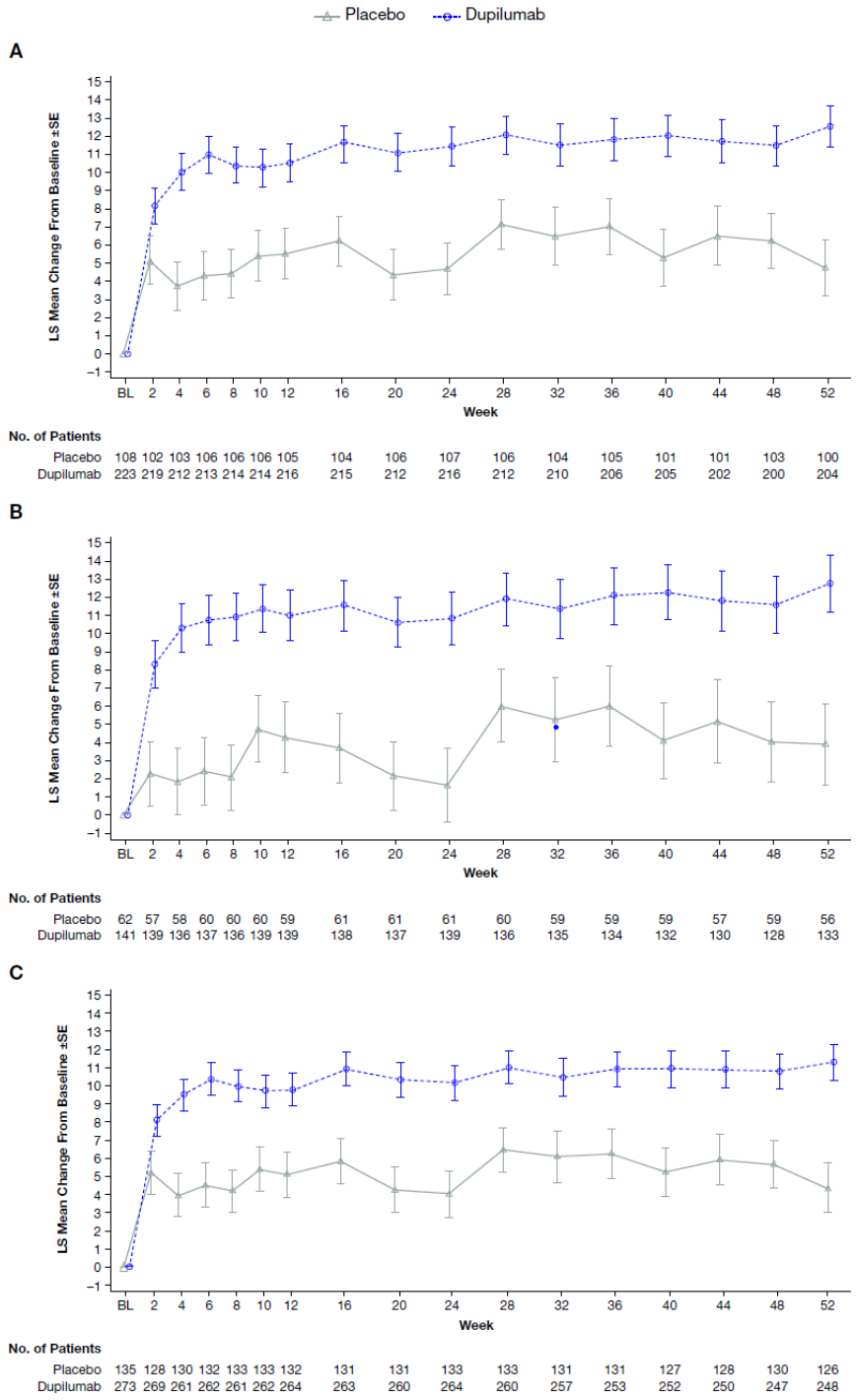
FE_{NO} denotes fractional exhaled nitric oxide, and ppb parts per billion

Figure S7. Kaplan–Meier Plot of Time to First Loss of Asthma Control Event During the 52-Week Intervention Period in the Populations with the Type 2 Inflammatory Asthma Phenotype (A) and with Blood Eosinophils ≥300 cells/µl (B).



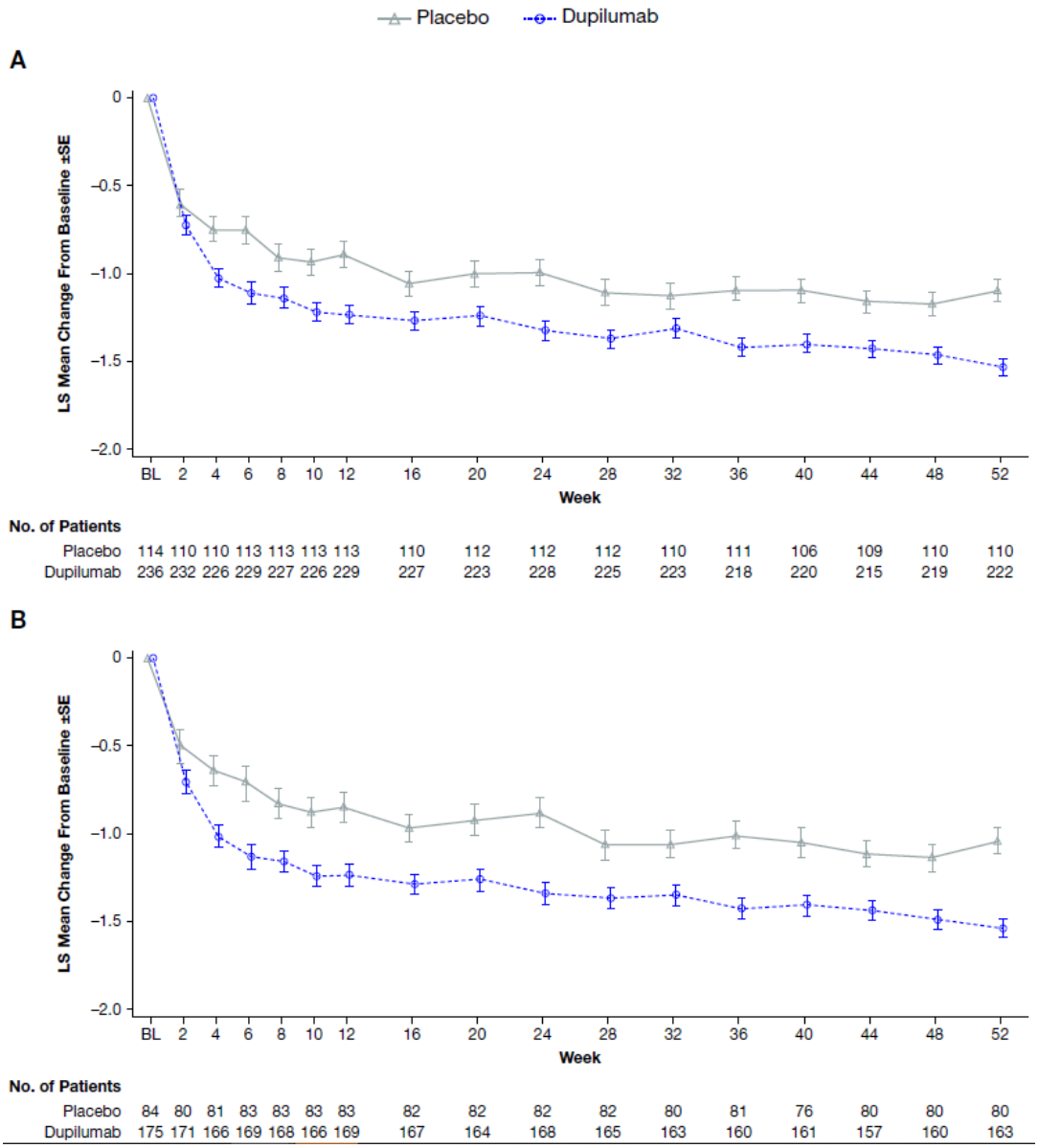
LOAC is defined as ≥6 additional reliever puffs of salbutamol/albuterol or levosalbutamol/levalbuterol in a 24-hour period (compared to baseline) on 2 consecutive days; increase in ICS dose ≥4 times the dose at visit 2; and a decrease in morning or evening peak flow of 30% or more on 2 consecutive days of treatment, based on the defined stability limit. The Treatment Period stability limit is defined as the respective mean morning or evening peak expiratory flow obtained over the last 7 days prior to randomization (day 1); a severe exacerbation event. FE_{NO} denotes fractional exhaled nitric oxide, ICS inhaled corticosteroid, LOAC loss of asthma control, and ppb parts per billion.

Figure S8. Change from Baseline Over the 52-Week Intervention Period in Percentage of Predicted Pre-Bronchodilator FEV_1 in Patients with Blood Eosinophils ≥150 cells/µl (A), with Baseline FE_{NO} ≥20 ppb (B), and in the Overall Intention-to-Treat Population (C).



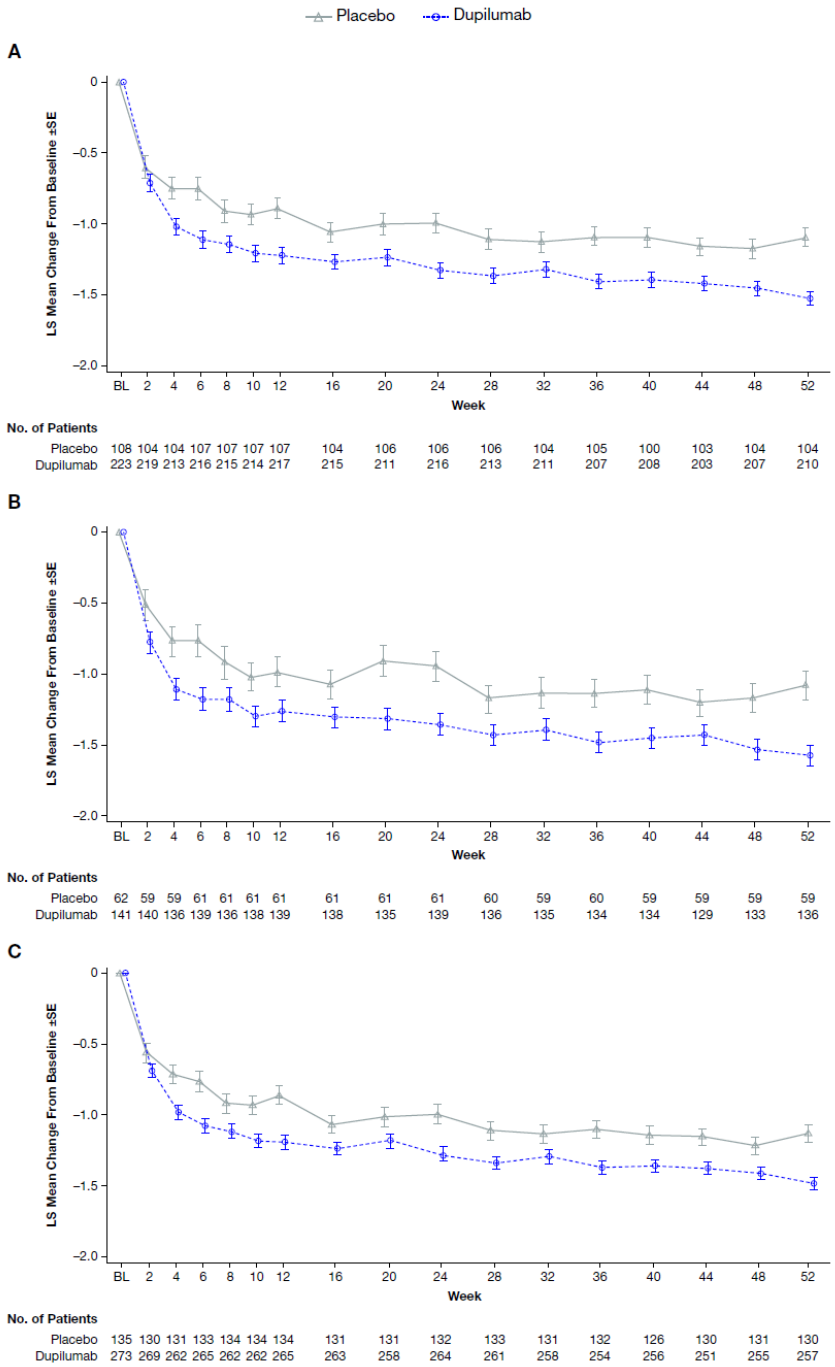
BL denotes baseline, FE_{NO} fractional exhaled nitric oxide, FEV_1 forced expiratory volume in 1 second, ITT intention-to-treat, LS least squares, and SE standard error.

Figure S9. Change from Baseline Over the 52-Week Intervention Period in ACQ-7-IA in Patients with the Type 2 Inflammatory Asthma Phenotype (A), with Baseline Eosinophils ≥300 cells/µl (B).



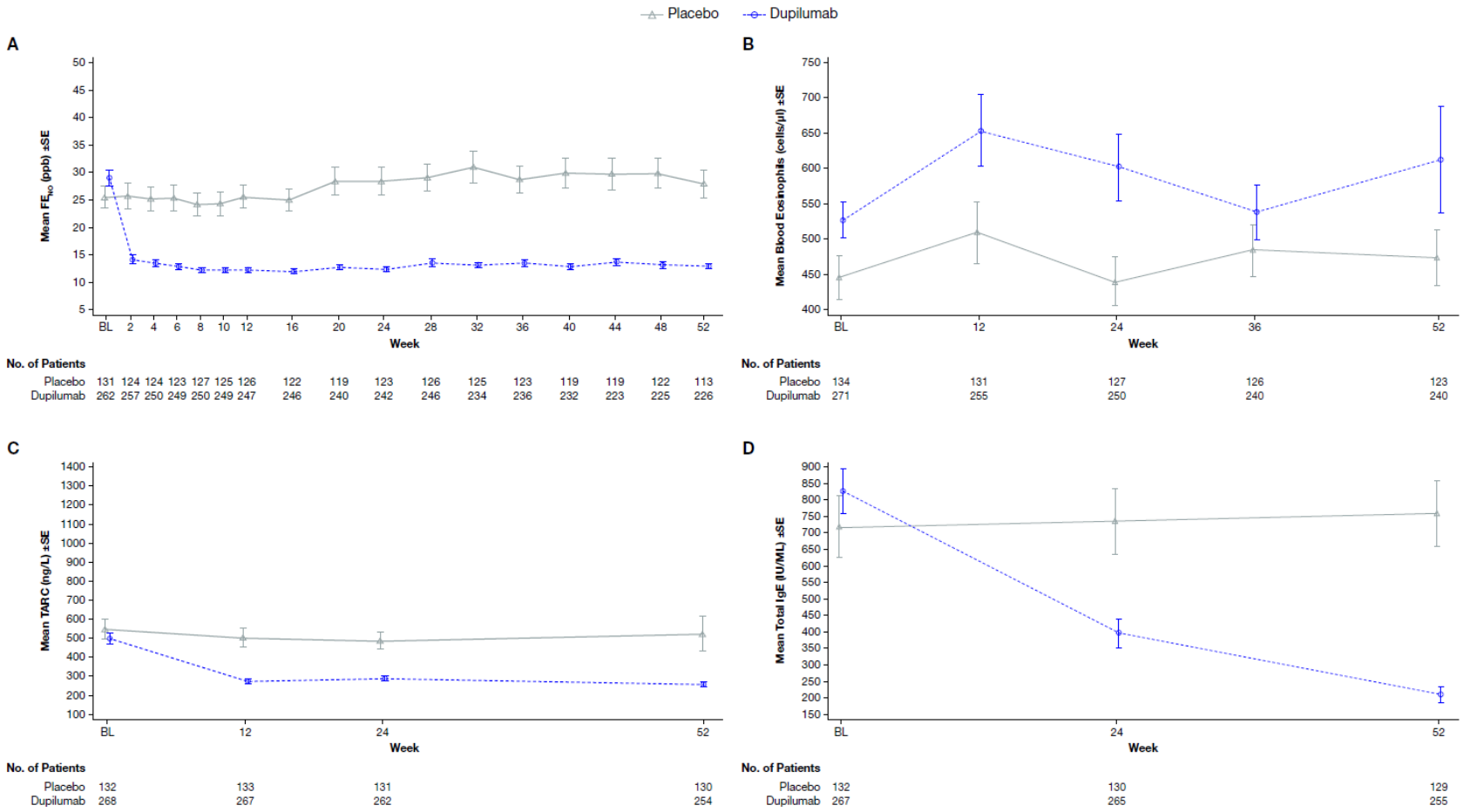
ACQ-7-IA denotes 7-item Asthma Control Questionnaire – Interviewer-Administered, BL baseline, FE_{NO} fractional exhaled nitric oxide, ITT intention-to-treat, LS least squares, and SE standard error.

Figure S10. Change from Baseline Over the 52-Week Intervention Period in ACQ-7-IA in Patients with Baseline Eosinophils ≥150 cells/µl (A), with Baseline FeNO ≥20 ppb (B), and in the Overall ITT Population (C).



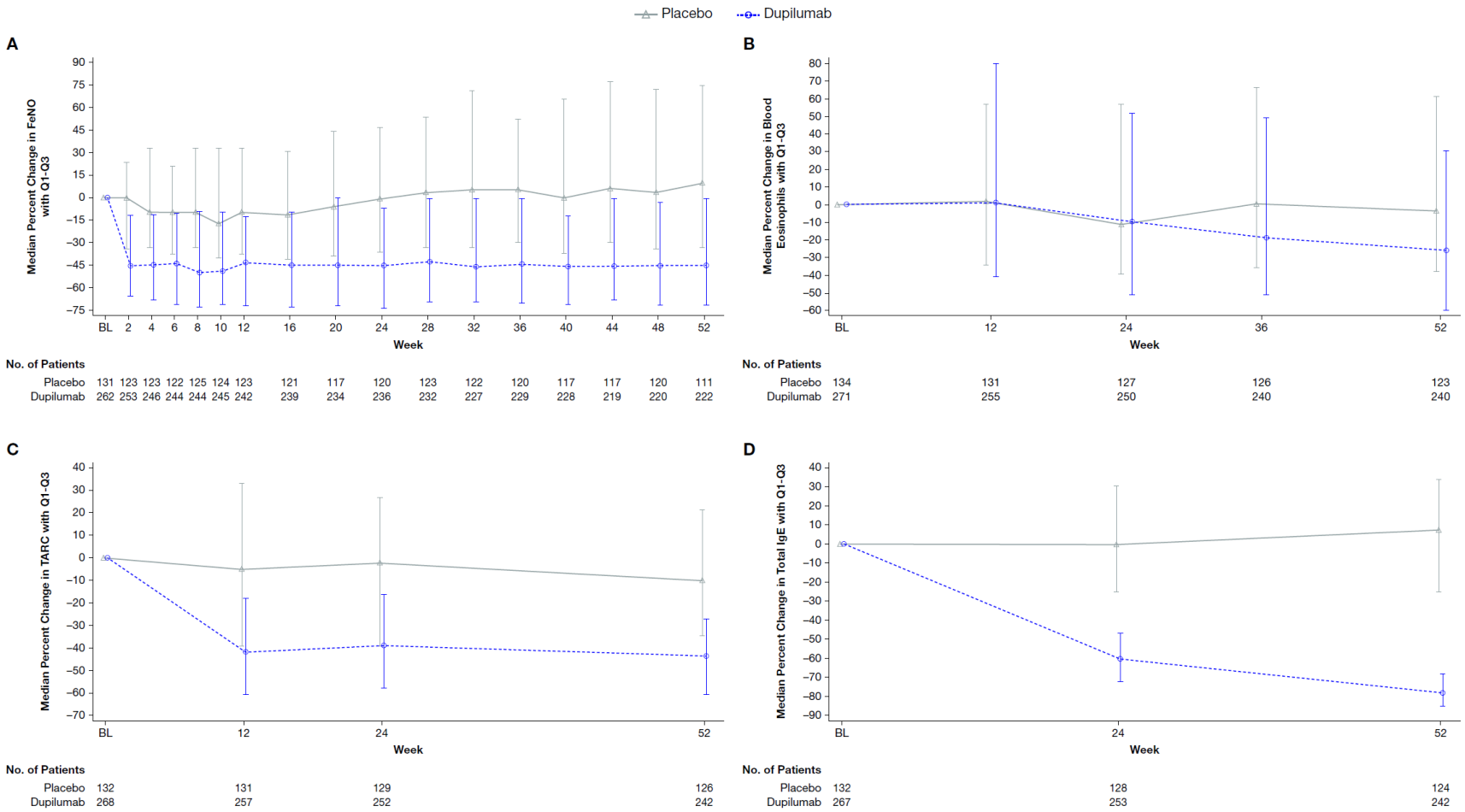
ACQ-7-IA denotes 7-item Asthma Control Questionnaire – Interviewer-Administered, BL baseline, FE_{NO} fractional exhaled nitric oxide, ITT intention-to-treat, LS least squares, and SE standard error.

Figure S11. Mean FE_{NO} (A), Blood Eosinophils (B), TARC (C), and Total IgE (D) in the Overall Safety Population.



BL denotes baseline, FE_{NO} fractional exhaled nitric oxide, IgE immunoglobulin E, and TARC serum thymus and activation-regulated chemokine.

Figure S12. Median Percent Change from Baseline in Type 2 Biomarkers 52-Week Intervention Period in the Overall Safety Population.



Only baseline and on-treatment post-baseline values (from first IMP to end of treatment or last IMP +14 days) are included.
BL denotes baseline, FE_{NO} fractional exhaled nitric oxide, IgE immunoglobulin E, and TARC serum thymus and activation-regulated chemokine.

Table S1. Low-, Medium-, and High-dose ICS in Children Aged 6 to >12 years

ICS	Total Daily Dose (µg)		
	Low	**Medium**	**High**
Beclometasone dipropionate (CFC)	100 to 200	>200 to 400	>400
Beclometasone dipropionate (HFA)	50 to 100	>100 to 200	>200
Budesonide (DPI)	100 to 200	>200 to 400	>400
Budesonide (HFA)	100 to 200	>200 to 400	>400
Budesonide (nebules)	250 to 500	>500 to 1000	>1000
Ciclesonide (HFA)	80	80 to 160	>160
Flunisolide (HFA)	160	>160 to <320	320
Fluticasone propionate (DPI)	100 to 200	>200 to 400	>400
Fluticasone propionate (HFA)	100 to 200	>200 to 500	>500
Mometasone furoate	110	≥220 to <440	≥440
Triamcinolone acetonide	400 to 800	>800 to 1200	>1200

*Protocol deviation; †Adverse events are reported at the Preferred Term level of the MedDRA hierarchy; ‡ Eosinophilia is defined as eosinophil count ≥3000 cells; CFC denotes chlorofluorocarbon propellant, DPI dry powder inhaler, HFA hydrofluoroalkane propellant, nebules nebulized solution, and ICS inhaled corticosteroid. Adapted from Global Initiative for Asthma (GINA) 2015 guidelines[1] with the addition of HFA and Flunisolide.

Table S2: Summary of Study End Points in Hierarchical Order*

	US Hierarchy					EU Hierarchy				
Order	**End points**	**Population**	**Placebo (N)**	**Dupilumab (N)**	**Difference vs Placebo**	**End points**	**Population**	**Placebo (N)**	**Dupilumab (N)**	**Difference vs Placebo**
1st	Exacerbations	Eos ≥300 cells/µl	0.665 (84)	0.235 (175)	0.353 (<.0001)	Exacerbations	Type 2	0.748 (114)	0.305 (236)	0.407 (<.0001)
2nd	Exacerbations	Eos ≥150 cells/µl	0.816 (108)	0.318 (223)	0.390 (<.0001)	Exacerbations	Eos ≥150 cells/µl	0.816 (108)	0.318 (223)	0.390 (<.0001)
3rd	Exacerbations	Type 2	0.748 (114)	0.305 (236)	0.407 (<.0001)	Exacerbations	Eos ≥300 cells/µl	0.665 (84)	0.235 (175)	0.353 (<.0001)
4th	FEV_1 pp week 12	Eos ≥300 cells/µl	4.83 (80)	10.15 (168)	5.32 (0.0036)	FEV_1pp week 12	Type 2	5.32 (110)	10.53 (229)	5.21 (0.0009)
5th	FEV_1pp week 12	Eos ≥150 cells/µl	5.65 (104)	10.63 (216)	4.98 (0.0020)	FEV_1pp week 12	Eos ≥150 cells/µl	5.65 (104)	10.63 (216)	4.98 (0.0020)
6th	FEV_1pp week 12	Type 2	5.32 (110)	10.53 (229)	5.21 (0.0009)	FEV_1pp week 12	Eos ≥300 cells/µl	4.83 (80)	10.15 (168)	5.32 (0.0036)
7th	ACQ-7-IA week 24	Eos ≥300 cells/µl	-0.88 (80)	-1.34 (166)	-0.46 (<.0001)	ACQ-7-IA week 24	Type 2	-1.00 (110)	-1.33 (227)	-0.33 (0.0001)
8th	ACQ-7-IA week 24	Eos ≥150 cells/µl	-0.98 (104)	-1.33 (214)	-0.36 (<.0001)	ACQ-7-IA week 24	Eos ≥150 cells/µl	-0.98 (104)	-1.33 (214)	-0.36 (<.0001)
9th	ACQ-7-IA week 24	Type 2	-1.00 (110)	-1.33 (227)	-0.33 (0.0001)	ACQ-7-IA week 24	Eos ≥300 cells/µl	-0.88 (80)	-1.34 (166)	-0.46 (<.0001)
10th	Exacerbations	FE_{NO} ≥20 ppb	0.705 (62)	0.271 (141)	0.384 (0.0004)	Exacerbations	FE_{NO} ≥20 ppb	0.705 (62)	0.271 (141)	0.384 (0.0004)
11th	FEV_1pp week 12	FE_{NO} ≥20 ppb	4.62 (62)	11.36 (141)	6.74 (0.0018)	FEV_1pp week 12	FE_{NO} ≥20 ppb	4.62 (62)	11.36 (141)	6.74 (0.0018)
12th	ACQ-7-IA week 24	FE_{NO} ≥20 ppb	-0.96 (62)	-1.37 (141)	-0.41 (0.0006)	ACQ-7-IA week 24	FE_{NO} ≥20 ppb	-0.96 (62)	-1.37 (141)	-0.41 (0.0006)
13th	Exacerbations	Full ITT	0.608 (135)	0.278 (273)	0.458 (<.0001)	Exacerbations	Full ITT	0.608 (135)	0.278 (273)	0.458 (<.0001)
14th	FEV_1pp week 12	Full ITT	5.17 (131)	9.85 (264)	4.68 (0.0012)	FEV_1pp week 12	Full ITT	5.17 (131)	9.85 (264)	4.68 (0.0012)
15th	Exacerbations	Eos≥300 & high ICS	0.697 (41)	0.251 (74)	0.360 (0.0019)	Exacerbations	Type 2 & high ICS	1.145 (50)	0.420 (102)	0.367 (0.0004)
16th	Exacerbations	Eos≥150 & high ICS	1.393 (47)	0.449 (94)	0.322 (0.0001)	Exacerbations	Eos≥150 & high ICS	1.393 (47)	0.449 (94)	0.322 (0.0001)
17th	Exacerbations	Type 2 & high ICS	1.145 (50)	0.420 (102)	0.367 (0.0004)	Exacerbations	Eos≥300 & high ICS	0.697 (41)	0.251 (74)	0.360 (0.0019)
18th	Exacerbations	FE_{NO}≥20 & high ICS	1.045 (28)	0.374 (64)	0.358 (0.0060)	Exacerbations	FE_{NO}≥20 & high ICS	1.045 (28)	0.374 (64)	0.358 (0.0060)
19th	Exacerbations	Full ITT & high ICS	0.844 (60)	0.362 (120)	0.429 (0.0021)	Exacerbations	Full ITT & high ICS	0.844 (60)	0.362 (120)	0.429 (0.0021)
20th	FE_{NO} week 12	Eos ≥ 300 cells/µl	-0.81 (80)	-21.40 (165)	-20.59 (<.0001)	FE_{NO} week 12	Type 2	-1.13 (110)	-18.97 (226)	-17.84 (<.0001)
21st	FE_{NO} week 12	Eos ≥150 cells/µl	-0.88 (104)	-18.76 (213)	-17.88 (<.0001)	FE_{NO} week 12	Eos ≥150 cells/µl	-0.88 (104)	-18.76 (213)	-17.88 (<.0001)
22nd	FE_{NO} week 12	Type 2	-1.13 (110)	-18.97 (226)	-17.84 (<.0001)	FE_{NO} week 12	Eos ≥300 cells/µl	-0.81 (80)	-21.40 (165)	-20.59 (<.0001)
23rd	FE_{NO} week 12	FE_{NO} ≥20 ppb	-2.49 (62)	-30.50 (140)	-28.00 (<.0001)	FE_{NO} week 12	FE_{NO} ≥20 ppb	-2.49 (62)	-30.50 (140)	-28.00 (<.0001)
24th	FE_{NO} week 12	Full ITT	-1.33 (130)	-16.09 (261)	-14.76 (<.0001)	FE_{NO} week 12	Full ITT	-1.33 (130)	-16.09 (261)	-14.76 (<.0001)

*Change from baseline in percentage of the predicted pre-bronchodilator FEV_1 at week 12 was the only key secondary end point. End points have been controlled for multiplicity. Eos denotes eosinophils, FE_{NO} fractional exhaled nitric oxide, ICS inhaled corticosteroid, ITT intention-to-treat, and ppb parts per billion.

Table S3. Sensitivity Analysis. Results for Primary and Key Secondary Analyses

	Annualized rate of severe exacerbations			Change from baseline in percentage of predicted pre-bronchodilator FEV1 at Week 12		
	No. of patients			No. of patients		
	Placebo	Dupilumab	Relative Risk vs placebo (95% CI)	Placebo	Dupilumab	LS Mean Difference vs placebo (95%CI)
Type 2 inflammatory asthma phenotype						
Primary Analysis	114	236	0.407 (0.274–0.605)	110	229	5.21 (2.14–8.27)
On-treatment analysis	114	233	0.403 (0.269–0.603)	110	227	5.20 (2.15–8.26)
PMM-MI	114	236	0.411 (0.277–0.609)	110	230	5.18 (2.13–8.23)
Control based PMM-MI	114	236	0.440 (0.293–0.661)	110	230	5.25 (2.18–8.31)
Patients with Baseline Eosinophils ≥300 cells/µl						
Primary Analysis	84	175	0.353 (0.222–0.562)	80	168	5.32 (1.76–8.88)
On-treatment analysis	84	172	0.351 (0.219–0.563)	80	166	5.22 (1.66–8.79)
PMM-MI	84	175	0.360 (0.227–0.569)	80	169	5.20 (1.65–8.74)
Control based PMM-MI	84	175	0.393 (0.244–0.632)	80	169	5.22 (1.65–8.79)

CI confidence interval, PMM-MI Pattern mixture model-multiple imputation

Table S4. Sensitivity Analysis. Tipping Point Analyses - Relative risk (p-value) for annualized event rate of severe exacerbation during 52-week treatment period

	Deflation Ratio on the Monthly Odds[b] for Early-study-withdrawal Patients in Placebo					
Inflation Ratio on the Monthly Odds[a] for Early-study-withdrawal Patients in Dupilumab	1	0.8	0.5	0.25	0.1	0.05
Type 2 inflammatory asthma phenotype						
1	0.411 (<.0001)	0.412 (<.0001)	0.414 (<.0001)	0.417 (<.0001)	0.417 (<.0001)	0.417 (<.0001)
2	0.426 (<.0001)	0.427 (<.0001)	0.429 (<.0001)	0.432 (<.0001)	0.433 (<.0001)	0.433 (<.0001)
5	0.459 (<.0001)	0.460 (<.0001)	0.463 (0.0001)	0.466 (0.0001)	0.466 (0.0001)	0.466 (0.0001)
10	0.496 (0.0005)	0.497 (0.0006)	0.500 (0.0006)	0.503 (0.0006)	0.504 (0.0007)	0.504 (0.0007)
20	0.539 (0.0030)	0.540 (0.0031)	0.544 (0.0033)	0.547 (0.0035)	0.548 (0.0036)	0.548 (0.0036)
50	0.600 (0.0173)	0.601 (0.0177)	0.605 (0.0192)	0.609 (0.0204)	0.610 (0.0208)	0.610 (0.0208)
100	0.645 (0.0472)	0.647 (0.0483)	0.651 (0.0517)	0.655 (0.0544)	0.656 (0.0555)	0.656 (0.0555)
Patients with Baseline Eosinophils ≥300 cells/µl						
1	0.360 (<.0001)	0.362 (<.0001)	0.364 (<.0001)	0.365 (<.0001)	0.367 (<.0001)	0.368 (<.0001)
2	0.374 (<.0001)	0.376 (<.0001)	0.378 (<.0001)	0.380 (<.0001)	0.382 (<.0001)	0.382 (<.0001)
5	0.404 (0.0001)	0.406 (0.0001)	0.409 (0.0001)	0.410 (0.0002)	0.413 (0.0002)	0.413 (0.0002)
10	0.440 (0.0006)	0.443 (0.0007)	0.445 (0.0007)	0.447 (0.0008)	0.450 (0.0008)	0.450 (0.0008)
20	0.488 (0.0034)	0.491 (0.0037)	0.494 (0.0040)	0.496 (0.0041)	0.499 (0.0044)	0.499 (0.0045)
50	0.558 (0.0235)	0.561 (0.0249)	0.564 (0.0267)	0.567 (0.0275)	0.570 (0.0288)	0.571 (0.0294)
100	0.608 (0.0614)	0.612 (0.0644)	0.615 (0.0682)	0.618 (0.0706)	0.622 (0.0735)	0.623 (0.0745)

[a]For each month (28 days), unobserved binary severe exacerbation event was imputed 40 times using a logistic regression imputation model, where the monthly binary odds for the missing cohort of dupilumab arm was predicted via inflating the observed odds with the inflation ratio assigned in this table

[b]For each month (28 days), unobserved binary severe exacerbation event was imputed 40 times using a logistic regression imputation model, where the monthly binary odds for the missing cohort of placebo arm was predicted via deflating the observed odds with the deflation ratio assigned in this table

Table S5. Sensitivity Analysis. Tipping Point Analyses - LS mean difference (P-value) based on tipping point analysis of change from baseline in prebronchodilator % predicted FEV1 at Week 12

The decreased value on the imputed value in Dupilumab group(%)[a]	The increased value on the imputed value in Placebo group (%)[b]									
	1	2	3	4	5	6	7	8	9	10
Type 2 inflammatory asthma phenotype										
-1	5.11 (0.0011)	5.08 (0.0012)	5.05 (0.0012)	5.02 (0.0013)	4.99 (0.0014)	4.96 (0.0015)	4.93 (0.0016)	4.91 (0.0017)	4.88 (0.0018)	4.85 (0.0019)
-2	5.08 (0.0012)	5.05 (0.0013)	5.02 (0.0013)	4.99 (0.0014)	4.96 (0.0015)	4.93 (0.0016)	4.90 (0.0017)	4.87 (0.0018)	4.84 (0.0020)	4.81 (0.0021)
-3	5.04 (0.0013)	5.01 (0.0014)	4.99 (0.0015)	4.96 (0.0016)	4.93 (0.0017)	4.90 (0.0018)	4.87 (0.0019)	4.84 (0.0020)	4.81 (0.0021)	4.78 (0.0023)
-4	5.01 (0.0014)	4.98 (0.0015)	4.95 (0.0016)	4.92 (0.0017)	4.89 (0.0018)	4.86 (0.0019)	4.84 (0.0020)	4.81 (0.0022)	4.78 (0.0023)	4.75 (0.0025)
-5	4.98 (0.0015)	4.95 (0.0016)	4.92 (0.0017)	4.89 (0.0018)	4.86 (0.0020)	4.83 (0.0021)	4.80 (0.0022)	4.77 (0.0024)	4.74 (0.0025)	4.71 (0.0027)
-6	4.95 (0.0017)	4.92 (0.0018)	4.89 (0.0019)	4.86 (0.0020)	4.83 (0.0021)	4.80 (0.0023)	4.77 (0.0024)	4.74 (0.0026)	4.71 (0.0027)	4.68 (0.0029)
-7	4.91 (0.0018)	4.88 (0.0019)	4.85 (0.0020)	4.82 (0.0022)	4.80 (0.0023)	4.77 (0.0025)	4.74 (0.0026)	4.71 (0.0028)	4.68 (0.0030)	4.65 (0.0031)
-8	4.88 (0.0020)	4.85 (0.0021)	4.82 (0.0022)	4.79 (0.0024)	4.76 (0.0025)	4.73 (0.0027)	4.70 (0.0028)	4.67 (0.0030)	4.65 (0.0032)	4.62 (0.0034)
-9	4.85 (0.0022)	4.82 (0.0023)	4.79 (0.0024)	4.76 (0.0026)	4.73 (0.0027)	4.70 (0.0029)	4.67 (0.0031)	4.64 (0.0033)	4.61 (0.0035)	4.58 (0.0037)
-10	4.81 (0.0023)	4.78 (0.0025)	4.75 (0.0026)	4.73 (0.0028)	4.70 (0.0030)	4.67 (0.0032)	4.64 (0.0034)	4.61 (0.0036)	4.58 (0.0038)	4.55 (0.0040)
Patients with Baseline Eosinophils ≥300 cells/µl										
-1	5.17 (0.0046)	5.13 (0.0049)	5.09 (0.0052)	5.05 (0.0056)	5.01 (0.0059)	4.97 (0.0063)	4.93 (0.0068)	4.90 (0.0072)	4.86 (0.0077)	4.82 (0.0082)
-2	5.13 (0.0050)	5.09 (0.0053)	5.05 (0.0056)	5.01 (0.0060)	4.97 (0.0064)	4.93 (0.0068)	4.89 (0.0073)	4.86 (0.0078)	4.82 (0.0083)	4.78 (0.0088)
-3	5.09 (0.0053)	5.05 (0.0057)	5.01 (0.0061)	4.97 (0.0065)	4.93 (0.0069)	4.89 (0.0073)	4.86 (0.0078)	4.82 (0.0083)	4.78 (0.0089)	4.74 (0.0095)
-4	5.05 (0.0058)	5.01 (0.0061)	4.97 (0.0065)	4.93 (0.0070)	4.89 (0.0074)	4.85 (0.0079)	4.82 (0.0084)	4.78 (0.0090)	4.74 (0.0096)	4.70 (0.0102)
-5	5.01 (0.0062)	4.97 (0.0066)	4.93 (0.0071)	4.89 (0.0075)	4.85 (0.0080)	4.82 (0.0085)	4.78 (0.0091)	4.74 (0.0097)	4.70 (0.0103)	4.66 (0.0109)
-6	4.97 (0.0067)	4.93 (0.0072)	4.89 (0.0076)	4.85 (0.0081)	4.81 (0.0086)	4.78 (0.0092)	4.74 (0.0098)	4.70 (0.0104)	4.66 (0.0111)	4.62 (0.0118)
-7	4.93 (0.0073)	4.89 (0.0077)	4.85 (0.0082)	4.81 (0.0087)	4.78 (0.0093)	4.74 (0.0099)	4.70 (0.0105)	4.66 (0.0112)	4.62 (0.0119)	4.58 (0.0127)
-8	4.89 (0.0079)	4.85 (0.0084)	4.81 (0.0089)	4.77 (0.0094)	4.74 (0.0100)	4.70 (0.0107)	4.66 (0.0113)	4.62 (0.0120)	4.58 (0.0128)	4.54 (0.0136)
-9	4.85 (0.0085)	4.81 (0.0090)	4.77 (0.0096)	4.74 (0.0102)	4.70 (0.0108)	4.66 (0.0115)	4.62 (0.0122)	4.58 (0.0130)	4.54 (0.0138)	4.50 (0.0146)
-10	4.81 (0.0092)	4.77 (0.0098)	4.74 (0.0104)	4.70 (0.0110)	4.66 (0.0117)	4.62 (0.0124)	4.58 (0.0131)	4.54 (0.0140)	4.50 (0.0148)	4.46 (0.0157)

[a]Imputed FEV1 values in dupilumab groups are subtracted by the shifting value. Negative imputed values are replaced with 0.
[b]Imputed FEV1 values in placebo group are increased by the shifting value.

Table S6. Breakdown of Patient Populations by Baseline Biomarkers Levels and in the Total Population

	Placebo (N=135)	**Dupilumab (N=273)**	**All (N=408)**
Randomized population, no. (%)	135 (100)	273 (100)	408 (100)
Efficacy populations, no. (%)			
Total population	135 (100)	273 (100)	408 (100)
Type 2 inflammatory asthma phenotype	114 (84.4)	236 (86.4)	350 (85.8)
Baseline blood eosinophils ≥300 cells/µL	84 (62.2)	175 (64.1)	259 (63.5)
Baseline blood eosinophils ≥150 cells/µL	108 (80.0)	223 (81.7)	331 (81.1)
Baseline FeNO ≥20 ppb	62 (45.9)	141 (51.6)	203 (49.8)
Safety population, no. (%)	134 (99.3)	271 (99.3)	405 (99.3)

Table S7. Baseline Demographics and Clinical Characteristics in the Population with the Type 2 Inflammatory Asthma Phenotype and in the Population with Baseline Blood Eosinophils ≥300 cells/µl

	Type 2 Inflammatory Phenotype		Baseline Blood Eosinophils ≥300 cells/µl		Total population	
Characteristic	**Placebo N = 114**	**Dupilumab N = 236**	**Placebo N = 84**	**Dupilumab N = 175**	**Placebo N = 135**	**Dupilumab N = 273**
Age, mean (SD) — yr	9.0 (1.6)	8.9 (1.6)	9.0 (1.5)	8.9 (1.6)	8.9 (1.6)	8.9 (1.7)
Male — no. (%)	78 (68.4)	152 (64.4)	58 (69.0)	116 (66.3)	87 (64.4%)	175 (64.1%)
Race — no. (%)						
White	102 (89.5)	208 (88.1)	75 (89.3)	151 (86.3)	118 (87.4%)	242 (88.6%)
Black	5 (4.4)	9 (3.8)	5 (6.0)	8 (4.6)	9 (6.7%)	11 (4.0%)
Asian	0	2 (0.8)	0	2 (1.1)	0	2 (0.7%)
American Indian or Alaska Native	0	1 (0.4)	0	1 (0.6)	0	1 (0.4%)
Other	7 (6.1)	16 (6.8)	4 (4.8)	13 (7.4)	8 (5.9%)	17 (6.2%)
Ethnicity – no. (%)						
Hispanic or Latino	51 (44.7)	104 (44.1)	35 (41.7)	111 (42.9)	60 (44.4%)	118 (43.2%)
Not Hispanic or Latino	63 (55.3)	132 (55.9)	49 (58.3)	148 (57.1)	75 (55.6%)	155 (56.8%)
Weight, mean (SD) — kg	37.08 (11.6)	35.60 (10.0)	36.97 (11.7)	35.45 (10.0)	36.94 (12.29)	35.39 (10.20)
Time since first diagnosis of asthma, mean (SD) — yr	5.23 (2.63)	5.78 (2.67)	5.54 (2.65)	5.78 (2.60)	5.19 (2.62)	5.75 (2.60)
Age at onset of asthma, mean (SD) — yr	3.8 (2.6)	3.2 (2.5)	3.5 (2.6)	3.2 (2.6)	3.8 (2.5)	3.2 (2.5)
Use of inhaled glucocorticoid — no. (%)						
High-dose ICS	50 (43.9)	102 (43.2)	41 (48.8)	74 (42.3)	60 (44.4)	120 (44.0)
Medium-dose ICS	64 (56.1)	131 (55.5)	43 (51.2)	98 (56.0)	75 (55.6)	150 (54.9)
ICS-only	1 (0.9)	8 (3.4)	1 (1.2)	5 (2.9)	1 (0.7)	8 (2.9)
ICS/LABA	101 (88.6%)	196 (83.1%)	72 (85.7%)	145 (82.9%)	119 (88.1)	231 (84.6)
ICS/other	12 (10.5%)	30 (12.7%)	11 (13.1%)	23 (13.1%)	15 (11.1)	32 (11.7)

Severe asthma exacerbations in past year, mean (SD) — no.	2.18 (1.55)	2.61 (2.58)	2.37 (1.71)	2.78 (2.90)	2.19 (1.51)	2.56 (2.43)
Number of exacerbations in past year — no. (%)						
1	47 (41.2)	85 (36.0)	32 (38.1)	64 (36.6)	53 (39.3%)	96 (35.2%)
2	32 (28.1)	75 (31.8)	20 (23.8)	53 (30.3)	42 (31.1%)	87 (31.9%)
3	21 (18.4)	29 (12.3)	19 (22.6)	17 (9.7)	23 (17.0%)	38 (13.9%)
≥4	14 (12.3)	47 (19.9)	13 (15.5)	41 (23.4)	17 (12.6%)	52 (19.0%)
With atopic medical condition* — no. (%)	103 (90.4)	226 (95.8)	79 (94.0)	171 (97.7)	107 (79.3)	240 (87.9)
With atopic dermatitis	41 (36.0)	95 (40.3)	35 (41.7)	80 (45.7)	45 (33.3)	103 (37.7)
Pre-BD FEV1, mean (SD) — liters	1.53 (0.46)	1.48 (0.39)	1.52 (0.48)	1.45 (0.39)	1.51 (0.45)	1.47 (0.38)
Pre-BD FEV1, mean (SD) — % predicted	78.36 (14.51)	77.66 (14.38)	77.87 (15.19)	76.37 (14.60)	78.98 (14.74)	77.63 (14.72)
FEV1 reversibility, mean (SD) — liters	15.81 (16.38)	21.47 (21.37)	16.20 (15.76)	22.94 (23.23)	15.63 (16.33)	21.56 (22.43)
$FEF_{25\%-75\%}$, mean (SD) — l/sec	1.28 (0.53)	1.27 (0.54)	1.24 (0.54)	1.23 (0.52)	1.27 (0.54)	1.28 (0.55)
FVC, mean (SD) — liters	2.08 (0.57)	2.00 (0.48)	2.09 (0.61)	1.98 (0.48)	2.05 (0.56)	1.98 (0.48)
Morning PEF, mean (SD) — l/min	187.96 (57.49)	196.35 (67.63)	186.57 (59.12)	192.32 (67.62)	189.40 (57.64)	195.48 (67.70)
Evening PEF, mean (SD) — l/min	198.72 (57.53)	203.27 (65.78)	197.06 (59.70)	199.97 (65.44)	199.55 (56.99)	202.63 (65.77)
Baseline ACQ-5-IA score, mean (SD)	2.15 (0.84)	2.18 (0.79)	2.17 (0.86)	2.18 (0.83)	2.16 (0.85)	2.17 (0.82)
ACQ-7-IA score, mean (SD)	2.12 (0.76)	2.15 (0.70)	2.15 (0.77)	2.16 (0.73)	2.11 (0.75)	2.14 (0.72)
Global PAQLQ(S)-IA score, mean (SD)	4.92 (1.13)	4.95 (1.08)	4.89 (1.09)	4.96 (1.13)	4.85 (1.18)	4.95 (1.10)
Morning asthma symptom score, mean (SD)	0.90 (0.72)	0.90 (0.78)	0.93 (0.73)	0.94 (0.79)	0.93 (0.74)	0.89 (0.79)
Evening asthma symptom score, mean (SD)	0.92 (0.72)	0.92 (0.77)	0.91 (0.72)	0.88 (0.79)	0.97 (0.72)	0.92 (0.77)
Blood eosinophil count, mean (SD) — cells/µl	520 (360)	600 (390)	640 (350)	740 (360)	450 (370)	503 (410)
Blood eosinophil count <150 cells/µl — no (%)	6 (5.3)	13 (5.5)	0	0	27 (20.0%)	50 (18.3%)
Blood eosinophil count, ≥150 cells/µl — no (%)	108 (94.7)	223 (94.5)	84 (100)	175 (100)	108 (80.0)	223 (81.7)

Blood eosinophil count, median (Q1:Q3) — cells/µl	440 (280:660)	510 (290:780)	530 (420:730)	660 (480:910)	400 (210:610)	470 (210:730)
Total IgE, median (Q1:Q3) — IU/ml	397.0 (144.0:862.0)	530.0 (213.0:1268.0)	554.0 (215.0:1101.0)	599.5 (331.0:1470.0)	348.0 (75.0:762.0)	441.0 (125.0:1080.0)
FE_{NO}, mean (SD) — ppb	28.38 (23.44)	31.83 (24.85)	31.33 (23.52)	34.53 (25.83)	25.44 (22.55)	28.83 (24.42)
FE_{NO} — ≥20 ppb	62 (54.4)	141 (59.7)	52 (61.2)	114 (65.1)	62 (45.9)	141 (51.6)
TARC, mean (SD) — pg/ml	567.96 (666.29)	525.58 (511.75)	579.27 (728.47)	573.85 (573.84)	546.55 (624.35)	499.43 (484.09)

*A patient is considered to have ongoing atopic medical condition if he/she has any of the following ongoing conditions: atopic dermatitis, allergic conjunctivitis, allergic rhinitis, eosinophilic esophagitis, food allergy, hives; or has baseline total IgE ≥100 IU/ml and at least one aeroallergen specific IgE is positive (≥ 0.35 IU/ml) at baseline.
ACQ-5-IA denotes 5-item Asthma Control Questionnaire — Interviewer-Administered, ACQ-7-IA 7-item Asthma Control Questionnaire — Interviewer-Administered, BD bronchodilator, $FEF_{25\%-75\%}$ forced expiratory flow 25%–75%, FE_{NO} fractional exhaled nitric oxide, FEV_1 forced expiratory volume in 1 second, FVC forced vital capacity, IgE immunoglobulin E, PAQLQ(S)-IA pediatric quality of life questionnaire (standardized) — Interviewer-Administered, PEF peak expiratory flow, ppb parts per billion, SD standard deviation, and TARC serum thymus and activation-regulated chemokine.

Table S8. Systemic Corticosteroid Exposure During the 52-Week Treatment Period in the population with the Type 2 Inflammatory Asthma Phenotype and in the Population with Eosinophils ≥300 cells/µl

	Type 2 Inflammatory Phenotype		Baseline Eosinophils ≥300 cells/µl	
	Placebo (N = 114)	**Dupilumab (N = 236)**	**Placebo (N = 84)**	**Dupilumab (N = 175)**
Patients with SCS use				
Yes — no. (%)	46 (40.4)	57 (24.2)	35 (41.7)	39 (22.3)
No — no. (%)	68 (59.6)	179 (75.8)	49 (58.3)	136 (77.7)
Individual annualized total SCS courses in all patients*†				
Mean (SD)	0.86 (1.32)	0.40 (0.90)	0.92 (1.40)	0.35 (0.81)
Individual annualized total SCS duration in all patients				
Mean (SD) — days	4.44 (7.29)	2.08 (5.44)	4.78 (7.87)	1.80 (4.62)
Adjusted annualized total SCS courses‡				
Estimate (95% CI)	0.860 (0.616, 1.200)	0.350 (0.256, 0.477)	0.806 (0.563, 1.154)	0.274 (0.188, 0.399)
Relative risk vs. placebo (95% CI)		0.407 (0.272, 0.609)		0.340 (0.212, 0.545)

*The total dose/number of SCS days for each patient divided by the number of patient-years followed in the treatment period for that patient.

†A course of SCS is considered continuous if treatment is separated by less than 7 days.

‡For all patients enrolled in the given population (either the baseline eosinophils ≥150 cells/µl or FE_{NO} ≥20 ppb population or the baseline eosinophils ≥300 cells/µl population).

CI denotes confidence interval, FE_{NO} fractional exhaled nitric oxide, ppb parts per billion, SCS systemic corticosteroid, and SD standard deviation.

Table S9. Change from Baseline in Lung Function Measurements Over the 52-Week Treatment Period in the Population with the Type 2 Inflammatory Asthma Phenotype

	Type 2 Inflammatory Phenotype					
	Baseline		Week 2		Week 52	
	Placebo (N = 114)	Dupilumab (N = 236)	Placebo (N = 114)	Dupilumab (N = 236)	Placebo (N = 114)	Dupilumab (N = 236)
Pre-BD FEV_1 % predicted						
Pre-BD FEV_1, mean (SD) — % predicted	78.36 (14.51)	77.66 (14.38)				
LS mean (SE) change — % predicted			4.84 (1.29)	8.11 (0.95)	4.36 (1.50)	12.15 (1.10)
LS mean difference vs. matching placebo (95% CI)				3.27 (0.40, 6.14)		7.79 (4.36, 11.22)
Change from baseline in pre-BD FEV_1						
Pre-BD FEV_1, mean (SD) — liters	1.53 (0.46)	1.48 (0.39)				
LS mean (SE) change — liters			0.08 (0.03)	0.14 (0.02)	0.24 (0.03)	0.41 (0.02)
LS mean difference vs. matching placebo (95% CI)				0.06 (0.01, 0.12)		0.17 (0.09, 0.24)
Change from baseline in post-BD FEV_1 % predicted						
Post-BD FEV_1, mean (SD) — % predicted	89.66 (15.54)	92.36 (14.70)				
LS mean (SE) change — % predicted			0.61 (1.26)	1.51 (0.92)	−0.75 (1.48)	3.62 (1.08)
LS mean difference vs. matching placebo (95% CI)				0.91 (−1.92, 3.73)		4.37 (0.95, 7.78)

Change from baseline in post-BD FEV_1						
Post-BD FEV_1, mean (SD) — liters	1.74 (0.49)	1.75 (0.43)				
LS mean (SE) change — liters			0.01 (0.02)	0.02 (0.02)	0.18 (0.03)	0.26 (0.02)
LS mean difference vs. matching placebo (95% CI)				0.01 (−0.04, 0.06)		0.09 (0.02, 0.16)
Change from baseline in morning PEF						
Morning PEF, mean (SD)	187.96 (57.49)	196.35 (67.63)				
LS mean (SE) change — l/min			5.57 (2.66)	6.50 (2.00)	19.88 (5.16)	31.45 (3.69)
LS mean difference vs. matching placebo (95% CI)				0.93 (−4.94, 6.79)		11.57 (−0.58, 23.71)
Change from baseline in evening PEF						
Evening PEF, mean (SD)	198.72 (57.53)	203.27 (65.78)				
LS mean (SE) change — l/min			7.62 (2.94)	10.60 (2.20)	16.46 (5.20)	28.20 (3.73)
LS mean difference vs. matching placebo (95% CI)				2.98 (−3.49, 9.46)		11.74 (−0.44, 23.92)
Change from baseline in FVC						
FVC, mean (SD)	2.08 (0.57)	2.00 (0.48)				
LS mean (SE) change — liters			0.07 (0.02)	0.07 (0.02)	0.27 (0.03)	0.37 (0.02)
LS mean difference vs. matching placebo (95% CI)				0.01 (−0.05, 0.06)		0.10 (0.03, 0.17)
Change from baseline in $FEF_{25\%–75\%}$						

$FEF_{25\%-75\%}$, mean (SD)	1.28 (0.53)	1.27 (0.54)				
LS mean (SE) change† — liters			0.11 (0.05)	0.28 (0.04)	0.29 (0.06)	0.60 (0.04)
LS mean difference vs. matching placebo (95% CI)†				0.17 (0.07, 0.27)		0.30 (0.17, 0.44)

†Derived from MMRM model with change from baseline in $FEF_{25\%-75\%}$ or post-BD FEV_1 values up to week 52 as the response variable, and treatment, age, baseline weight group, region, sex, ethnicity, baseline height, baseline eosinophil level, baseline $F_{E_{NO}}$ level, baseline ICS dose level, visit, treatment-by-visit interaction, baseline $FEF_{25\%-75\%}$ value and baseline-by-visit interaction as covariates. Type 2 inflammatory asthma phenotype population is defined as the randomized patients with baseline blood eosinophils ≥ 150 cells/µl or baseline $F_{E_{NO}}$ ≥20 ppb.

CI denotes confidence interval, BD bronchodilator, $FEF_{25\%-75\%}$ forced expiratory flow 25%–75%, $F_{E_{NO}}$ fractional exhaled nitric oxide, FEV_1 forced expiratory volume in 1 second, FVC forced vital capacity, ICS inhaled corticosteroid, LS least squares, MMRM mixed-effects model with repeated measures, PEF peak expiratory flow, ppb parts per billion, SD standard deviation, and SE standard error.

Table S10. Change from Baseline in Lung Function Measurements Over the 52-Week Treatment Period in the Population with Baseline Eosinophils ≥300 cells/µl

	Baseline Eosinophils ≥300 cells/µl					
	Baseline		**Week 2**		**Week 52**	
	Placebo (N = 114)	**Dupilumab (N = 236)**	**Placebo (N = 114)**	**Dupilumab (N = 236)**	**Placebo (N = 114)**	**Dupilumab (N = 236)**
Pre-BD FEV_1 % predicted						
Pre-BD FEV_1, mean (SD) — % predicted	77.87 (15.19)	76.37 (14.60)				
LS mean (SE) change from baseline — % predicted			3.94 (1.51)	7.49 (1.08)	4.08 (1.80)	12.43 (1.30)
LS mean difference vs. matching placebo (95% CI)				3.55 (0.09, 7.01)		8.35 (4.12, 12.57)
Change from baseline in pre-BD FEV_1						
Pre-BD FEV_1, mean (SD) — liters	1.52 (0.48)	1.45 (0.39)				
LS mean (SE) change from baseline — liters			0.07 (0.03)	0.13 (0.02)	0.25 (0.04)	0.42 (0.03)
LS mean difference vs. matching placebo (95% CI)				0.07 (−0.00, 0.13)		0.17 (0.09, 0.26)
Change from baseline in post-BD FEV_1 % predicted						
Post-BD FEV_1, mean (SD) — % predicted	89.55 (16.03)	91.55 (15.67)				

LS mean (SE) change — % predicted			0.10 (1.49)	0.52 (1.07)	−0.52 (1.79)	3.13 (1.30)
LS mean difference vs. matching placebo (95% CI)				0.41 (−3.04, 3.86)		3.65 (−0.58, 7.88)
Change from baseline in post-BD FEV_1						
Post-BD FEV_1, mean (SD) — liters	1.75 (0.53)	1.74 (0.44)				
LS mean (SE) change — liters			0.00 (0.03)	0.01 (0.02)	0.19 (0.04)	0.26 (0.03)
LS mean difference vs. matching placebo (95% CI)				0.00 (−0.06, 0.07)		0.07 (−0.01, 0.16)
Change from baseline in morning PEF						
Morning PEF, mean (SD)	185.57 (59.12)	192.32 (67.62)				
LS mean (SE) change — l/min			5.85 (2.97)	7.94 (2.21)	19.75 (6.33)	32.80 (4.49)
LS mean difference vs. matching placebo (95% CI)				2.08 (−4.76, 8.93)		13.05 (−2.03, 28.12)
Change from baseline in evening PEF						
Evening PEF, mean (SD)	197.06 (59.70)	199.97 (65.44)				
LS mean (SE) change — l/min			5.64 (3.42)	10.02 (2.53)	14.67 (6.39)	26.60 (4.54)
LS mean difference vs. matching placebo (95% CI)				4.38 (−3.52, 12.27)		11.93 (−3.25, 27.10)
Change from baseline in FVC						
FVC, mean (SD)	2.09 (0.61)	1.98 (0.48)				
LS mean (SE) change — liters			0.07 (0.03)	0.06 (0.02)	0.29 (0.04)	0.38 (0.03)

LS mean difference vs. matching placebo (95% CI)				−0.01 (−0.07, 0.05)		0.08 (−0.00, 0.17)
Change from baseline in $FEF_{25\%-75\%}$						
$FEF_{25\%-75\%}$, mean (SD)	1.24 (0.54)	1.23 (0.52)				
LS mean (SE) change† — liters			0.07 (0.05)	0.29 (0.04)	0.26 (0.07)	0.65 (0.05)
LS mean difference vs. matching placebo (95% CI)†				0.22 (0.10, 0.33)		0.39 (0.23, 0.55)

†Derived from MMRM model with change from baseline in $FEF_{25\%-75\%}$ % or post-BD FEV_1 values up to week 52 as the response variable, and treatment, age, baseline weight group, region, sex, ethnicity, baseline height, baseline eosinophil level, baseline FE_{NO} level, baseline ICS dose level, visit, treatment-by-visit interaction, baseline $FEF_{25\%-75\%}$ value and baseline-by-visit interaction as covariates. Type 2 inflammatory asthma phenotype population is defined as the randomized patients with baseline blood eosinophils ≥150 cells/µl or baseline FE_{NO} ≥20 ppb.

CI denotes confidence interval, BD bronchodilator, $FEF_{25\%-75\%}$ forced expiratory flow 25%−75%, FE_{NO} fractional exhaled nitric oxide, FEV_1 forced expiratory volume in 1 second, FVC forced vital capacity, ICS inhaled corticosteroid, LS least squares, MMRM mixed-effects model with repeated measures, PEF peak expiratory flow, SD standard deviation, and SE standard error.

Table S11. Change from Baseline in Patients Reported Outcomes Over the 52-Week Treatment Period in the Population with the Type 2 Inflammatory Asthma Phenotype

	Type 2 Inflammatory Phenotype					
	Baseline		**Week 24**		**Week 52**	
	Placebo (N = 114)	**Dupilumab (N = 236)**	**Placebo (N = 114)**	**Dupilumab (N = 236)**	**Placebo (N = 114)**	**Dupilumab (N = 236)**
Change from baseline in ACQ-7-IA						
ACQ-7-IA, mean (SD)	2.12 (0.76)	2.15 (0.70)	-			
LS mean (SE) change			−0.99 (0.07)	−1.32 (0.05)	−1.09 (0.06)	−1.53 (0.05)
LS mean difference vs. matching placebo (95% CI)				−0.33 (−0.50, −0.16)		−0.44 (−0.59, −0.30)
Change from baseline in PAQLQ(S)-IA*						
PAQLQ(S)-IA, mean (SD)	4.92 (1.13)	4.95 (1.08)				
LS mean (SE) change			1.11 (0.09)	1.30 (0.07)	1.19 (0.08	1.53 (0.06)
LS mean difference versus matching placebo (95% CI)				0.19 (−0.03, 0.40)		0.34 (0.16, 0.52)
Change from baseline in morning asthma symptom score						
Asthma morning symptom score, mean (SD)	0.90 (0.72)	0.90 (0.78)				
LS mean (SE) change			−0.46 (0.05)	−0.56 (0.04)	−0.50 (0.05)	−0.61 (0.04)

LS mean difference vs. matching placebo (95% CI)				−0.10 (−0.22, 0.02)		−0.12 (−0.23, 0.00)
Change from baseline in evening asthma symptom score						
Asthma evening symptom score, mean (SD)	0.92 (0.72)	0.92 (0.77)				
LS mean (SE) change			−0.44 (0.06)	−0.53 (0.04)	−0.50 (0.06)	−0.59 (0.04)
LS mean difference vs. matching placebo (95% CI)				−0.09 (−0.22, 0.04)		−0.09 (−0.22, 0.03)
Change from baseline in nocturnal awakenings						
Nocturnal awakenings, mean (SD)	0.32 (0.45)	0.39 (0.68)				
LS mean (SE) change			−0.23 (0.03)	−0.29 (0.02)	−0.26 (0.03)	−0.32 (0.02)
LS mean difference vs. matching placebo (95% CI)				−0.06 (−0.14, 0.02)		−0.06 (−0.12, −0.00)

*Derived from MMRM model with change from baseline in PAQLQ(S)-IA global score values up to week 52 as the response variable, and treatment, age, baseline weight group, region, baseline eosinophil level, baseline FE_{NO} level, baseline ICS dose level, visit, treatment-by-visit interaction, baseline PAQLQ(S)-IA global score value, and baseline-by-visit interaction as covariates.

ACQ-7-IA denotes 7-item Asthma Control Questionnaire − Interviewer-Administered, CI confidence interval, FE_{NO} fractional exhaled nitric oxide, ICS inhaled corticosteroid, LS least squares, MMRM mixed-effects model with repeated measures, PAQLQ(S)-IA pediatric quality of life questionnaire (standardized) − Interviewer-Administered, SD standard deviation, and SE standard error.

Table S12. Change from Baseline in Patients Reported Outcomes Over the 52-Week Treatment Period in the Population with Baseline Eosinophils ≥300 cells/µl

	Baseline Eosinophils ≥300 cells/µl					
	Baseline		**Week 24**		**Week 52**	
	Placebo (N = 114)	**Dupilumab (N = 236)**	**Placebo (N = 114)**	**Dupilumab (N = 236)**	**Placebo (N = 114)**	**Dupilumab (N = 236)**
Change from baseline in ACQ-7-IA						
ACQ-7-IA, mean (SD)	2.15 (0.77)	2.16 (0.73)				
LS mean (SE) change			−0.88 (0.09)	−1.34 (0.06)	−1.04 (0.08)	−1.54 (0.06)
LS mean difference vs. matching placebo (95% CI)				−0.46 (−0.66, −0.26)		−0.50 (−0.68, −0.32)
Change from baseline in PAQLQ(S)-IA*						
PAQLQ(S)-IA, mean (SD)	4.89 (1.09)	4.96 (1.13)				
LS mean (SE) change			1.06 (0.10)	1.36 (0.08)	1.23 (0.09)	1.56 (0.07)
LS mean difference vs. matching placebo (95% CI)				0.30 (0.06, 0.54)		0.33 (0.12, 0.53)
Change from baseline in morning asthma symptom score						
Asthma morning symptom score, mean (SD)	0.93 (0.73)	0.94 (0.79)				
LS mean (SE) change			−0.45 (0.06)	−0.55 (0.04)	−0.50 (0.06)	−0.62 (0.04)

LS mean difference vs. matching placebo (95% CI)				−0.10 (−0.24, 0.04)		−0.12 (−0.26, 0.02)
Change from baseline in evening asthma symptom score						
Asthma morning symptom score, mean (SD)	0.91 (0.72)	0.88 (0.79)				
LS mean (SE) change			−0.42 (0.06)	−0.53 (0.05)	−0.51 (0.06)	−0.60 (0.04)
LS mean difference vs. matching placebo (95% CI)				−0.11 (-0.26, 0.03)		−0.09 (−0.24, 0.05)
Change from baseline in nocturnal awakenings						
Nocturnal awakenings, mean (SD)	0.34 (0.47)	0.37 (0.68)				
Baseline, mean (SD)			−0.21 (0.04)	−0.28 (0.03)	−0.26 (0.03)	−0.32 (0.02)
LS mean (SE) change				−0.06 (−0.16, 0.03)		−0.06 (−0.13, 0.01)

*Derived from MMRM model with change from baseline in PAQLQ(S)-IA global score values up to week 52 as the response variable, and treatment, age, baseline weight group, region, baseline eosinophil level, baseline FE_{NO} level, baseline ICS dose level, visit, treatment-by-visit interaction, baseline PAQLQ(S)-IA global score value and baseline-by-visit interaction as covariates.

ACQ-7-IA denotes 7-item Asthma Control Questionnaire – Interviewer-Administered, CI confidence interval, FE_{NO} fractional exhaled nitric oxide, ICS inhaled corticosteroid, LS least squares, MMRM mixed-effects model with repeated measures, PAQLQ(S)-IA pediatric quality of life questionnaire (standardized) – Interviewer-Administered, SD standard deviation, and SE standard error.

Table S13. Change from Baseline in FeNO to Week 12 in the Type 2 Inflammatory Asthma Phenotype Population and Baseline Eosinophils ≥300 cells/µl Population

	Type 2 Inflammatory Phenotype		**Baseline Blood Eosinophils ≥300 cells/µl**	
	Placebo	**Dupilumab**	**Placebo**	**Dupilumab**
FE_{NO} (ppb)	**N = 114**	**N = 236**	**N = 84**	**N = 175**
Baseline				
Mean (SD)	28.38 (23.44)	31.83 (24.85)	31.33 (23.52)	34.53 (25.83)
Week 12				
Mean (SD)	28.77 (24.26)	12.40 (6.78)	31.08 (25.10)	12.58 (6.76)
LS Mean change from Baseline (SE)*	-1.13 (1.43)	-18.97 (1.04)	-0.81 (1.69)	-21.40 (1.21)
LS Mean difference vs Placebo (95% CI)*		-17.84 (-21.05, -14.63)		-20.59 (-24.60, -16.59)
P value vs Placebo*		<.0001		<.0001

*Derived from MMRM model with change from baseline in FeNO up to Week 12 as the response variable, and treatment, age, weight group, region, baseline eosinophil level, baseline ICS dose level, visit, treatment by-visit interaction, baseline FeNO value and baseline-by-visit interaction as covariates.

CI denotes confidence interval, FE_{NO} fractional exhaled nitric oxide, ICS inhaled corticosteroid, LS least squares, MMRM mixed-effects model with repeated measures, SD standard deviation, and SE standard error.

Table S14. Summary of Status in the Anti-Drug Antibody Assay – ADA population

Event, no. patients (%)	Placebo (N = 133)	Dupilumab (N = 269)
ADA-negative	129 (97.0)	252 (93.7)
ADA-positive	4 (3.0)	17 (6.3)
Persisten response	1 (0.8)	9 (3.3)
Transient response	2 (1.5)	5 (1.9)
Inderterminate response	1 (0.8)	3 (1.1)
Peak post-baseline titer		
Low (<1000)	3 (2.3)	16 (5.9)
Moderate (1000–10000)	1 (0.8)	1 (0.4)
High (>10000)	0	0

ADA, anti-drug antibodies.

ADA responses had no meaningful effect on efficacy and safety

5 References

1. Global Initiative for Asthma (GINA). Global Strategy for Asthma Management and Prevention. 2015. Available from: http://ginasthma.org/wpcontent/uploads/2016/01/GINA_Report_2015_Aug11-1.pdf. Accessed March 2018.
2. Juniper EF, Svensson K, Mörk AC, Ståhl E. Measurement properties and interpretation of three shortened versions of the asthma control questionnaire. Respir Med 2005;99:553-8.
3. Juniper EF, Buist AS, Cox FM, Ferrie PJ, King DR. Validation of a standardized version of the Asthma Quality of Life Questionnaire. Chest 1999;115:1265-70.
4. Castro M, Corren J, Pavord ID, et al. Dupilumab efficacy and safety in moderate-to-severe uncontrolled asthma. N Engl J Med 2018;378:2486-96.
5. Fan C, Zhang D, Zhang C. Robust small-sample inference for fixed effects in general Gaussian linear models. J Biopharm Stat 2012;22: 544-564.
6. Harville DA. Maximum likelihood approaches to variance component estimation and to related problems. J Am Stat Assoc 1977;72:320–338.